Not All Cops Are Bayesian: Selection Neglect in Data-Driven Policing

Juan B. González Santiago De Martini Ervyn Norza Santiago M. Perez-Vincent *

August 10, 2026

(Most Recent Version Here)

Abstract

Policing is increasingly data-driven, but crime data is a selected sample that reflects where crime is observed, not where it happens. We run a lab-in-the-field experiment testing how senior police officers in Colombia infer crime from selected data. Officers take crime data at face value, turning differences in observability into biased predictions. This failure reflects officers' mental models of crime data, not missing information. Because crime predictions guide enforcement, selection neglect can sustain inaccurate statistical discrimination purely from bad inference on selected data.

*González: Department of Economics, University of Southern California, juanbgon@usc.edu. De Martini: Department of Economics, University of Southern California, sd59576@usc.edu. Norza: National Police of Colombia, eh.norza10@uniandes.edu.co. Perez-Vincent: Inter American Development Bank, santiagoper@iadb.org. We are grateful to the Center for Effective Global Action (CEGA) for generously funding this study and to the Police Postgraduate School (ESPOL) in Bogotá for facilitating the experiments. We thank Giorgio Coricelli, Cary Frydman, Felipe Gonçalves, Jonathan Libgober, Alejandro Martínez-Marquina, Paulina Oliva, and the participants of the Berlin PhD Conference in Economics, LAX at UCLA, and BEEC at Caltech and UCSB for thoughtful feedback on the study.

I Introduction

Policing is increasingly data-driven, but the crime data it relies on is a selected sample of crime. Consider inferring the incidence of theft in the Bogotá localities of Barrios Unidos and Tunjuelito. A data-driven officer who uses crime reports, the official crime data, would infer Barrios Unidos has twice the rate of theft. However, these areas have the same incidence of theft according to representative victimization surveys. Why does crime data differ so much from crime incidence? The answer lies in crime *observability*: a victim of crime in Barrios Unidos is twice as likely to report the incident to the authorities and thus make it a crime data observation. Selection bias is pervasive in most crime datasets —citizen reports, calls to 911, police-filed field interviews—, as what incidents become observations depends on selective reporting and enforcement. Crime data shows where crime is observed, not where it happens, so taking crime data at face value can turn differences in observability into biased crime predictions.

Enforcement is not only guided by crime data; it often affects what incidents become crime data. Enforcement induces selective monitoring: crimes are more likely to be recorded in places and groups receiving more patrols, stops, or searches (Chen et al., 2023; Feigenberg & Miller, 2022). If officers take those records at face value, more enforcement would lead to more predicted crime, which can then guide more enforcement. As a result, selection neglect can sustain policing disparities through inaccurate crime predictions (Hübert & Little, 2023). In the United States, where minorities are more intensively monitored and thus overrepresented in crime data (Gonçalves et al., 2025; Pierson et al., 2020), neglecting selection can produce statistical discrimination even in the absence of racial prejudice. Despite these implications, whether officers actually neglect selection in crime data remains untested.

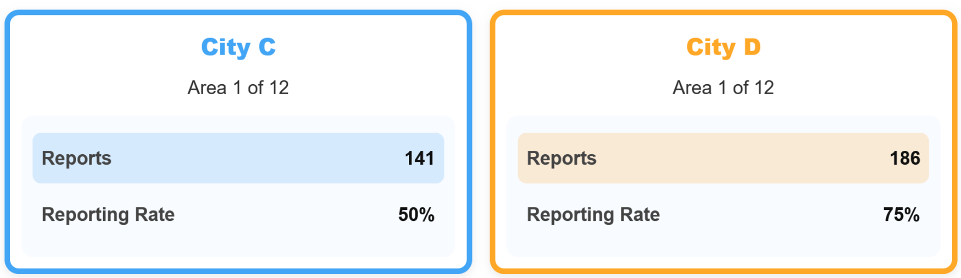
Colombia provides an ideal setting to address this question. Because the National Police relies on crime reports filed by citizens, selection there is driven by citizen reporting rather than enforcement, making it transparent, measurable, and exogenous to any individual officer's decisions. We leverage national and local victimization surveys to measure selective reporting (DANE, 2023; EPV, 2025), and document that reporting rates vary systematically across types of crime, victim characteristics, and areas. Despite crime reports being selected, officers are instructed to use them to assign patrols (PNVCC, 2010), without any instructions to adjust for selection bias. To validate whether patrols in fact track this selected data, we use high-frequency GPS data covering every patrol unit in eight Bogotá *cuadrantes* (police jurisdictions) in June 2025. We find they do: one additional crime report in a 300-meter cell is associated with roughly a third more patrol time. Moreover, our participating senior officers report relying on crime reports above any other source when making enforcement decisions. Together, police doctrine, survey, and administrative data suggest that officers act on a selected signal.

Although patrolling data can show the co-location of patrols and crime data, it cannot identify whether this co-location is caused by selection neglect. First, police records contain observed crime rather than underlying crime, and enforcement decisions rather than the predictions that motivate them. Thus, they lack a benchmark for the accuracy of crime predictions, and cannot separate these from other operational motives. Second, crime, enforcement, and observability are jointly determined (Ang et al., 2025; Gonçalves et al., 2026; Weisburd, 2021). Identifying selection neglect requires exogenous variation in reporting that leaves true crime, which is unlikely to exist in the field. Therefore, we directly elicit crime predictions from the officers who make them in the field, and create the identifying variation in an experiment.

In this paper, we conduct a lab-in-the-field experiment with senior officers of the National Police of Colombia. These officers, the universe of promoting Majors in the country, have on average 14 years of experience, and a quarter of them served as station commanders, using crime data to assign patrols. In the experiment, officers complete a series of crime prediction problems in which they observe crime data about two fictitious areas and are incentivized to correctly infer in which of them more crimes happened. In particular, officers see the number of *crime reports*, along with the *reporting rate* (the probability of a crime being reported). Because crimes are reported with different probabilities in each area, crime reports are a selected sample of underlying crime, and officers can take this selected sample at face value or adjust reports for selection by simply dividing them by the reporting rate.

Consider the example in Figure 1. An officer who takes reports at face value would infer more crime in the area on the right, but one that adjusts for the difference in observability would make the opposite prediction. By varying reporting rates and true crime independently each round, we generate the identifying variation the field cannot provide. This task not only matches the data available to officers in the field, but is, by design, the simplest version of predicting crime from selected data. Officers are provided with precise information about observability, the adjustment needed is a simple division, and they also have a calculator available on screen. In other words, this setting provides a conservative test of selection neglect.

Figure 1: Main Experimental Design



Our results show that officers neglect selection when predicting crime from selected data. We compare error rates across the two types of predictions generated by our design. When the area with more reports also has more true crime, so adjusting for selection does not change the correct prediction, officers err in less than one-fourth of decisions. This rate absorbs inattention and other errors unrelated to selection. In contrast, when adjustment changes the correct prediction (as in Figure 1), the error rate increases by 36 percentage points, making predictions significantly worse than chance. Because the only thing that changes between these two types of predictions is the need to adjust for selection, this excess prediction error is attributable to selection neglect. These errors are economically meaningful rather than confined to close comparisons: weighting each incorrect prediction by the true crime gap between areas, the implied loss averages more than eight crimes per decision, over half the average crime gap in the experiment. This excess error persists throughout the experiment, is pervasive across our sample, and is not explained by confusion about the task or limited arithmetic skills.

Neglect not only reduces accuracy, it biases predictions toward wherever crime is more observable. Holding crime constant, officers become more likely to infer more crime in an area as its reporting rate increases. Thus, officers turn differences in crime observability into biased crime predictions. We then test whether this pattern extends to the systematic differences in crime observability across groups documented in the literature. We introduce groups by bundling areas into two abstract cities, ruling out prejudice, and implement two conditions. In the baseline, reporting rates are identically distributed across cities; in the treatment condition, they are on average 30 percentage points higher in one city, creating selection bias across groups. The distribution of crime remains identical across cities and conditions, making any prediction bias towards a city entirely the result of inference error. We find that officers are equally likely to predict more crime in either city at baseline. In contrast, under selection bias across groups, officers are 20 percentage points more likely to predict more crime in the city where crime is more observable. These results show that disparities in crime beliefs can arise from bad inference on selected data without prejudice, consistent with prior theoretical work (Hübert & Little, 2023). Because minorities are overselected in crime data (Chen et al., 2023; Feigenberg & Miller, 2022; Gonçalves et al., 2025; Vomfell & Stewart, 2021), our results point to selection neglect as a cognitive mechanism that can feed policing disparities.

The disparity in crime predictions we document is not justified by any true crime difference, as both groups have equal distributions of crime by design. In other words, the belief disparity we identify is unambiguously *inaccurate*. This matters not only for potential downstream disparities, but also for the tests that identify these disparities and their source. Traditional outcome tests, originally proposed by Becker (1971), take accurate beliefs as the null hypothesis against which to test racial bias (Anwar & Fang, 2006; Knowles et al., 2001). Omitting the possibility of inaccurate beliefs can make analysts mistake poor inference for racial animus (Bohren et al., 2023). Our results question this assumption of outcome tests, and contribute direct belief evidence to recent work focusing on the role of inaccurate stereotypes and poor targeting in the justice system (Arnold et al., 2018; Feigenberg & Miller, 2022).

Why do these experts neglect selection even in the most favorable setting? Making the correct crime prediction in this setting involves two steps: (1) correctly *representing* the inference problem, that is, recognizing that crime reports are a selected sample of crime, and (2) conditional on that representation, correctly *computing* the adjustment. Several pieces of evidence point towards neglect being a failure of representation. First, when given the option to view reporting rates or not for making one prediction, over a third of officers choose to ignore it, consistent with this data not being demanded by their mental model of the problem. Importantly, information demand is linked to previous experience in the field: officers responsible for reading crime data and assigning patrols are significantly more likely to look at reporting rates than officers in administrative roles. Second, we probe mental models of crime data by asking officers a battery of questions about potential biases in crime data they routinely work with in the field. We find that officers who are more aware of the biases of selected data *in the field* are better at correcting selection bias *in the lab*. Third, most officers are able to correctly solve an equivalent problem not framed as a crime prediction —calculating the number of students in a class from incomplete observations— but then fail to implement this calculation when it is part of a crime prediction. This suggests officers bring to the task the mental models about crime data developed in the field. Fourth, we show that making reporting rates more salient does not affect neglect, suggesting mental models are entrenched and not developed ad hoc. In contrast, we find no evidence of computational frictions driving the results. Overall, selection neglect among officers seems a representational failure, with officers using incorrect mental models developed in the field to make crime predictions.

We further test these mechanisms through a randomized treatment. With the idea that giving officers the correct answer will only help if neglect arises from computation, but will not if it is a representational failure, we randomize officers into three groups. One-third receive advice from a *Correcting Algorithm* that adjusts for selection — so it recommends the correct prediction, another third gets it from a *Neglecting Algorithm* that takes reports at face value —so it gives the wrong advice in selection-sensitive decisions, and a control group who gets no advice. Officers are told which data their algorithm uses and are free to follow or override its advice. We find that officers override the advice that clashes with the inference rule they applied prior to the algorithm, and follow it when it confirms their predictions. As a result, even an algorithm that correctly computes the adjustment fails to improve crime predictions, but a naive algorithm makes them worse. These results, which are difficult to reconcile with salience or computational frictions as the drivers of neglect, illustrate the limits of interventions that often work in the lab when applied to experienced practitioners with entrenched mental models.

Related work

Policing increasingly relies on crime data to predict crime and guide policing decisions (Brayne, 2020; Perry et al., 2013), so biases in crime data can distort policing decisions. Prior work has addressed the consequences of selection bias in crime data, both for predictive algorithms trained on it (Arnold et al., 2025; Lum & Isaac, 2016) and for analysts trying to draw inferences from such datasets (Durlauf & Heckman, 2020; Gonçalves et al., 2025; Knox et al., 2020). However, how police officers infer crime from selected crime data remains unexplored. We directly elicit crime predictions to show that officers take selected crime data at face value. This paper thus contributes to a broader literature on how agents in the criminal justice system form (inaccurate) beliefs (Angelova et al., 2025; Arnold et al., 2018; Bhuller & Sigstad, 2024).

Because neglect turns differences in observability into biased predictions, it is a cognitive mechanism for statistical discrimination in policing. This paper joins recent work on the cognitive drivers of policing decisions, like reflection (Dube et al., 2024; Owens et al., 2018), implicit associations (Glaser, 2015), or cognitive depletion (Ferrazares, 2025). Selection neglect and these mechanisms are distinct from taste and stereotypes, and they generate disparities in crime predictions in the absence of racial prejudice. We provide direct evidence that these predictions can be largely *inaccurate*, even when made by senior officers in a simple setting. This finding questions outcome tests that benchmark racial bias against accurate beliefs (Anwar & Fang, 2006; Knowles et al., 2001), as formalized by Bohren et al. (2023).

Finally, this paper contributes to a broader literature identifying selection neglect. While some of this work comes from the field (Cruces et al., 2013; DellaVigna & Kaplan, 2007), most prior studies provide evidence from the lab (Ali et al., 2021; Enke, 2020; Esponda & Vespa, 2018; Jin et al., 2021). We complement these approaches by implementing a controlled environment with expert practitioners in the field. This matters for two reasons. First, external validity: we document that not only experimental subjects but experts show this cognitive failure, as Koehler and Mercer (2009) and Malmendier and Shanthikumar (2007) study among investors. The second one is arguably more important for research: practitioners have stable mental models, developed with experience, that lab subjects lack. We probe these mental models and find suggestive evidence that they drive selection neglect, as officers' mental representation of the inference problem determines the information they attend to and the computations they make (Bordalo et al., 2026; Esponda et al., 2024; Schwartzstein, 2014). As a consequence of these entrenched representations, interventions that are effective in the lab may not transfer to practitioners.

The rest of the paper proceeds as follows. Section II uses patrolling data to show that Colombian police officers use crime reports to assign patrols, documents the selection bias in this data, and discusses the limits of administrative data. Section III describes our sample of officers and the experimental design. Section IV presents the results on the prevalence of selection neglect, as well as its consequences for statistical discrimination. Section V explores the mechanisms driving neglect, using algorithmic advice to tell them apart. Section VI discusses the implications of our findings and their position within the prior literature and concludes.

II Setting

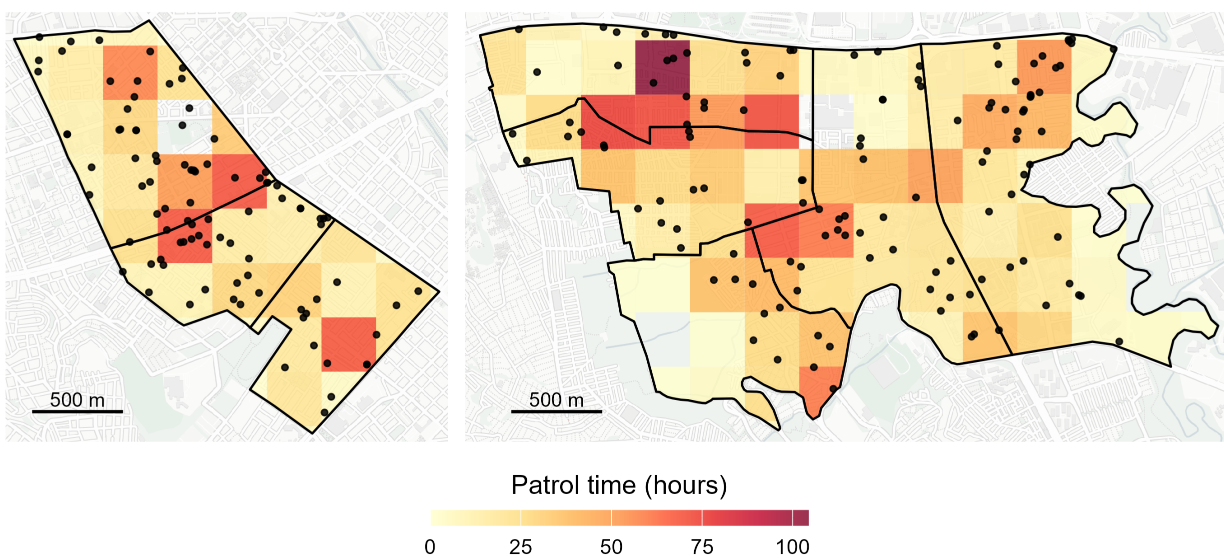
A Data-Driven Policing in Colombia

Policing in Colombia is explicitly data-driven. Crime reports have been a central input into policing decisions for decades, and since the early 2000s they have been consolidated in a centralized statistical system (SIEDCO). Since 2010, patrolling has been governed by the Plan Nacional de Vigilancia Comunitaria por Cuadrantes, which instructs commanders to use reported crime data to decide where and when to deploy patrols (PNVCC, 2010). Our own survey evidence suggests that officers have adapted to this data-driven regime: among our sample of senior officers, crime reports are considered the most relevant source of information to plan patrols and assign resources, above personal experience, live information from the ground, or even direct orders (see Figure A.2).

We can see this in administrative patrol data. We obtained high-frequency (20 seconds) GPS records of every patrol unit assigned to eight police *cuadrantes* in southern Bogotá, for June 2025, and matched them to the crime reports SIEDCO registered in the same area over the same period. We divide the jurisdictions into 300-meter cells and measure how long officers spent inside each one. The result is 150 cells, 218 reported crimes, and an average of seventeen hours of patrol per cell over the month. Figure 2 shows the distribution of crime reports and patrol time over the study area.

Figure 2: Patrol time and crime reports in eight *cuadrantes* of Bogotá, June 2025

(a) CAI Centenario Olaya **(b)** CAI Ismael Perdomo



Notes: Each tile is a 300m × 300m grid cell, shaded by the total time patrol units spent inside it during June 2025, measured from GPS pings weighted by the seconds elapsed until that officer's next fix (capped at 300 seconds). Black outlines are the official *cuadrantes*, the patrol beats to which officers are permanently assigned: three in Centenario and five in Perdomo. Black dots are individual crimes reported to SIEDCO (93 in Centenario, 127 in Perdomo). Tiles are clipped to the cuadrante boundaries for legibility, so the figure displays 123 of the 150 cells used in the analysis. Cells inside a boundary with no shading were never entered by a patrol unit during June, or contain the CAI station building and are excluded throughout. Basemap by OpenStreetMap and Carto contributors.

Patrols co-locate with reported crime. Table A.1 shows that cells with one additional reported crime in June received about six more hours of patrol that month, roughly a third more than the average cell ($p < 0.001$). The pattern holds at higher frequency: one additional report in the previous week is associated with about twenty extra minutes of patrol ($p < 0.001$). These comparisons hold the *cuadrante* fixed, so they compare cells patrolled by the same officers, who are permanently assigned to their *cuadrante*. The concentration of patrol time on reported crime is therefore not only the product of commanders sending more officers to busier areas: it is how officers distribute their time within the area they are responsible for. The co-location of crime reports and patrols comes mostly from variation across cells: including a cell fixed effect mutes the correlation, but this might be due to the short temporal coverage of our data.

These three sources of evidence —official strategy, survey responses, and patrol GPS data— establish crime reports as the central input into policing decisions. But this metric captures only the crimes that are reported. While reports can also be filed by police officers or prosecutors, in practice over 80% are filed by citizens (CEJ, 2025), either in person at a police station or through a dedicated online platform. Because reports are overwhelmingly citizen-filed, which crimes become observable to the police depends on citizen behavior. Not all victims file a report, and the National Police itself acknowledges that reported crime offers only a partial view of underlying crime (Rodríguez Cuadrado, 2008). The central challenge, however, is not that crime reports capture only a subset of crimes: it is that they are a *selected* sample of crime.

B Selection Bias in Crime Reports

According to national victimization surveys, only 28.8% of crimes are reported to the police (DANE, 2023). Reported crimes are not a random sample of all crimes: there are systematic differences in which crimes are reported and which victims report them. The 2025 Perception and Victimization Survey of Bogotá (EPV, 2025) provides rich data on reporting conditional on victimization. For reference, among over twenty thousand respondents, almost three thousand report being victims of at least one crime, for a total of 4,403 incidents. Table A.2 regresses whether each of these incidents was reported to the police on victim characteristics and crime type. Reporting varies sharply with both. Younger, higher-income, and more-educated victims are significantly more likely to report. Reporting also varies by crime type. Victims of theft — the most common crime — report in 43% of cases, whereas victims of sexual violence only do so in 27% of cases. Overall, not all crimes are equally observable, creating selection bias in crime data.

Reporting rates vary across space with the composition of crime types and victims, making crime more observable in some areas than others. As a result, the map of reported crime does not correspond to the map of crime. Consider the Bogotá localities of Barrios Unidos and Tunjuelito, whose theft data are depicted in Figure A.1. According to the EPV, which is representative at the locality level and measures victimization, the incidence of theft is essentially the same in both localities. Yet *reported* thefts —both according to the EPV and to administrative police data— are twice as high in Barrios Unidos. This asymmetry between crime reports and incidence is mainly driven by reporting behavior: residents of Barrios Unidos are almost twice as likely to report a theft. Figure A.3 shows these differences in reporting behavior across space are not confined to this example, but occur across all localities of Bogotá. Therefore, an officer who takes crime reports at face value might confound higher observability with higher incidence: crime data shows where crime is observed, not where it happens.

Importantly for this paper, selection bias in crime data in Colombia is quite transparent and not directly attributable to officers' short-run decisions. Selective reporting determines which crimes become observable, and this behavior is measurable through victimization surveys. Moreover, because officers do not determine the selection themselves, their inference is less likely to be confounded by motivated reasoning about their own conduct, as it might be where selection arises from officers' own stop or arrest decisions. These features make Colombia an unusually well-suited setting to study whether officers account for selection bias when making crime predictions.

C Do Officers Adjust for Selection Bias?

We have established that the National Police treats reported crime as its primary metric of crime, and this measure suffers from selection bias. Do crime predictions account for selection bias? The *cuadrante* doctrine does not consider the observability of crime, and the crime data systems of the National Police do not adjust for differential reporting rates. However, this does not settle whether officers themselves adjust when making crime predictions. Officers know their jurisdictions well, and may understand that some areas or some crimes generate fewer reports than others. On the other hand, they may not adjust because they lack the information to do so: reporting rates are not part of the police statistical system. The question is therefore open: does selection bias in crime reports translate into biased crime predictions, or do officers correct for it? And if officers do not correct for it, is it because they do not have the necessary information?

Why an Experiment

Despite Colombia being an ideal setting to test whether officers neglect selection bias in crime data, this mechanism cannot be identified from administrative patrol data alone. First, police records contain observed crime rather than underlying crime, and enforcement decisions rather than the predictions that motivate them. Thus, they lack a benchmark for the accuracy of crime predictions, and cannot separate these from other operational motives, especially as the objective function of policing is unknown (Stashko, 2023). Moreover, crime, enforcement, and observability are endogenously determined (Weisburd, 2021). Identifying selection neglect requires exogenous variation in reporting that leaves true crime unchanged. In practice, such variation is unlikely to be found in the field, as criminals respond to changes in observability (Gonçalves et al., 2026), and enforcement affects observability through direct monitoring (Chen et al., 2023; Feigenberg & Miller, 2022) and trust (Ang et al., 2025). For these reasons, we need an environment where the objective function is clear, predictions and true crime are observed, and where crime incidence and observability vary exogenously. Our experimental design puts senior police officers, the relevant decision makers, in such a setting.

III Design

A Participants

Our participants are captains of the National Police of Colombia. Captains usually serve as station commanders, reading crime data to predict crime across space and allocate patrols. In other words, these are the officers responsible for adjusting for selection bias, so they are the relevant sample to study selection neglect. In contrast, patrolling officers follow the orders of station commanders, and officers above captains work in strategic and operational decisions that do not involve direct allocation of patrols.

Procedures. — In September 2025, 202 Captains of the National Police of Colombia were attending the Police Postgraduate School (ESPOL) in Bogotá, completing the mandatory training required for promotion to Major. These officers constitute the universe of promoting Majors in 2025 and were instructed by superior officers to attend our experimental session as part of their training activities. Upon arrival, officers were invited to participate in a "Policing Science Study" and were offered an incentive of \$5 for participation, and up to \$14 in performance bonuses. Participation was voluntary and officers could decline without penalty.[1] Eighteen officers (8.9%) declined to participate before receiving any more information about the experiment. Of the 184 officers who began the study, all but two completed it, leaving 182 completers and no evidence of endogenous attrition. Of these, 27 fail a sanity check located at the beginning of the experiment and designed to flag disengagement. Because their responses through the experiment are indistinguishable from random choice, we focus on the other 155 participants for the analysis, and report the results with the entire sample in Appendix C. Officers were paid \$10.42 for 33 minutes of experiment, on average.

Final sample. — The final sample consists of 155 captains with substantial professional experience. Officers have an average of 14 years of service in the National Police of Colombia (minimum 12 years), and all hold at least an undergraduate degree, a requirement for promotion to this rank; 15% also hold a master's degree. Importantly for our purposes, participants have extensive experience making operational decisions: 25% were serving as station commanders immediately prior to the training, and an additional 40% held operative positions involving the allocation of police resources. In terms of demographics, 89% of participants are men, and ages range from 32 to 45 years (mean 36). Because our sample consists of experienced captains who are being promoted, it is, if anything, positively selected in terms of ability and experience.

B The Crime Prediction Task

The preregistered Crime Prediction Task is the core task in the experiment. In each decision, officers are presented with two abstract *areas* and then predict which one had more crime. Based on the presented information, the participants can calculate the number of crimes in each area, although this number remains unobserved to the participants.[2] We randomly select two predictions for payment, and officers receive \$5 for each prediction they got right. Note that areas are abstract and carry no social meaning:

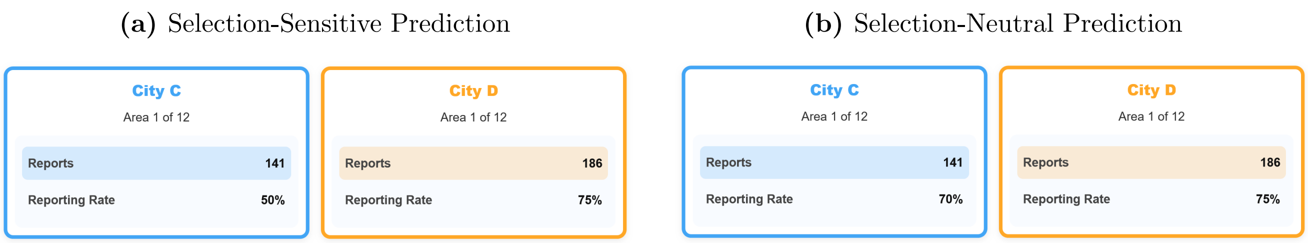
[1]The study received approval from the IRB at the University of Southern California (Protocol UP-25-00548), and all participants provided informed consent. The experimental design was preregistered before data collection at the AEA RCT Registry (AEARCTR-0016438).

[2]The data generating process of the number of crimes is calibrated to field data, so officers do not see unfamiliar figures. See Appendix B for more details of the DGP.

they are identified only by a label and a color, and each belongs to one of two abstract *cities*. Cities serve as stand-ins for any groups whose crime might differ —age, socioeconomic status, race— without invoking any of them. Because individuals and groups are abstract, officers hold no priors or stereotypes about them, and the design rules out taste-based discrimination by construction.

To predict which area had more crime, officers see two pieces of information. The first is the number of crime reports, the crime data used in the field. The second is the reporting rate: the probability that a crime in that area is reported, which measures the observability of crime. Given the data generating process, the best guess of the number of crimes is the number of reports *adjusted* by the reporting rate: $\frac{\#\text{Reports}}{\text{Reporting Rate}}$. We make sure officers understand how crime reports are generated through an extensive instruction period, including practice rounds and instruction comprehension questions. [3]

Figure 3: Screens of the Crime Prediction Task



Notes: The figure shows two examples of decision screens from the Crime Prediction Task. Panel (a) shows a selection-sensitive prediction, where adjusting for the observability of crime (reports/rate) changes the prediction. Panel (b) shows a selection-neutral prediction, where adjusting for observability or taking crime reports at face value lead to the same prediction.

Figure 3 illustrates the decision screen. Consider Panel (a): the area on the left-hand side received 141 reports, while the area on the right received 186. Taking reports at face value means predicting more crime in the area on the right. However, crime reports suffer from selection bias: crime is more observable in the area on the right. Adjusting for observability, the correct answer is that the area on the left had more expected crime ($141/0.5 = 282$) than the area on the right ($186/0.75 = 248$). Because adjusting for selection changes the optimal decision, these predictions identify who neglects selection. Note that adjusting for selection is deliberately simple. The optimal prediction requires one division, officers have an on-screen calculator and one minute per decision, and they complete comprehension and incentivized arithmetic questions before beginning.

Two features of the data-generating process make prediction errors interpretable. First, crime and reporting rates are drawn independently, and officers are told so explicitly. This matters: if officers believed that high-reporting areas also had more crime, following raw reports would be defensible inference rather than error. Independence removes that argument. Second, we fix the number of reports at exactly the reporting rate times the number of crimes, eliminating sampling noise. An officer who adjusts for selection is therefore always correct, so prediction errors reflect cognitive error rather than bad luck.

This design delivers the variation the field cannot provide: crime and observability (reporting rates) move independently across decisions, so we can change how selected the data is while holding true crime

[3]Instructions include examples such as: "Imagine that 200 crimes occurred in an area. If the reporting rate is 50%, we expect 100 crimes to be reported."

fixed. That variation generates two types of predictions, represented in Figure 3. Panel (a) shows a *selection-sensitive* decision, where adjusting for selection changes the resulting prediction: an officer who takes reports at face value necessarily predicts wrong. Panel (b) shows a *selection-neutral* decision, where the area with more reports is also the area with more crime, so adjusting for selection does not change the answer. The two are visually indistinguishable ex ante; telling them apart requires performing the adjustment. We take selection-neutral predictions as a benchmark, as they capture errors unrelated to failing to adjust for selection: inattention, confusion, randomizing, etc. The excess error rate in selection-sensitive decisions is our measure of selection neglect.

C Experimental Flow

Introduction — After listening to the instructions, officers complete several training rounds and answer nine comprehension questions. They then answer two incentivized questions that mirror the adjustment needed in the task, but presented as an arithmetic task. Before starting the task, participants can ask any clarifying questions.

Part I — At the beginning of the task, participants complete one *sanity check*, where they are only shown crime reports. In the absence of any other information, participants should predict the area with the higher number of reports. We use this decision to filter out subjects who are not engaged with the task, who comprise 16% of the total sample.[4] Then, all participants complete eight *low-salience* crime predictions, which include three decoy variables in addition to the number of reports and reporting rate.[5] We use these eight predictions to test whether making reporting rates less salient changes predictions. Finally, on the last round, all participants have to choose whether to see reporting rates or not for that last prediction. We take this *information choice* as evidence of whether they take reporting rates as an input in their inference process, but we do not use the corresponding prediction to calculate the prediction error in our results.

Main Task – Part II — Participants complete six rounds of the Crime Prediction Task with exactly the same setting described in Figure 3. Across these rounds, crime and reporting rates vary independently and within-subjects. These six rounds are the object of our main analysis on the prevalence of selection neglect among officers.

In these six predictions, we randomize participants into two between-subjects treatments. One half sees *Selection Bias Across Groups*: areas belonging to one city have higher reporting rates on average. The other half sees *No Selection Bias Across Groups*: both cities have the same distribution of reporting rates. Crucially, crime is equally distributed across cities in both treatments. We use these treatments to test whether officers carry selection bias across groups into biased predictions.

Finally, for the last six predictions, we randomize participants into three groups. The *Control* group continues making predictions as in the main task. The *Correcting Algorithm* group is provided with an

[4] 27 of the 182 participants failed this check. The check successfully flagged low engagement: their subsequent choices are statistically indistinguishable from chance and significantly faster than those from the main sample. Our primary analysis therefore excludes these participants. For transparency, Appendix C reports all main analyses including these participants.

[5] In particular, we include the average age, the unemployment rate, and the average socioeconomic status of each area. These variables do not provide any additional information conditional on the number of reports and reporting rates, so they act as uninformative decoys that make reporting rates less salient.

"algorithmic recommendation" that correctly adjusts for selection bias; participants are informed about this adjustment. The *Neglecting Algorithm* group is provided with a similar algorithmic recommendation, yet this one takes reports at face value and does not adjust for selection bias; participants are informed about this lack of adjustment. This between-subjects treatment serves to test the mechanisms behind neglect, and provides evidence about the limits of information provision.

One prediction from Part II is randomly chosen for payment. In total, participants complete 22 crime predictions across the two parts, of which we use 20 for our analysis.

End Questions — After completing the experiment, participants answer a short, unincentivized survey. Among these questions, officers report what sources of information are the most relevant for assigning patrols and allocating resources, as well as how aware they are about bias in the crime data they routinely work with in the field. In particular, we ask them whether crime reports capture spatial and temporal crime patterns, whether they are the best available data, whether citizens know how to report crimes, and whether reported data is generally biased. Because these responses are highly correlated, we follow Kling et al. (2007) and aggregate z-scores of each into an index of *Bias Awareness*, where higher values reflect a stronger belief that crime data in the field is biased. This measure allows us to relate experimental inference errors to officers' beliefs about real crime data. Participants also report their years of experience in the National Police of Colombia and the nature of their most recent position (administrative, operational, or surveillance). Officers in surveillance positions are those acting as station commanders and directly assigning patrols. Finally, participants provide basic demographic information.

Table 1: Summary of Experimental Design

Intro	Comprehension questions; arithmetic questions (up to $4)
Part I	Ten predictions (1 chosen for $5 if correct)
Prediction 1	Sanity check (prediction not used for analysis)
Predictions 2-9	Low-salience predictions (decoy information)
Prediction 10	Information choice (prediction not used for analysis)
Part II	Twelve crime predictions (1 chosen for $5 if correct)
Predictions 1-6	Main Task
Predictions 7-12	Control, Correcting Algorithm, or Neglecting Algorithm
End	Survey questions and demographics

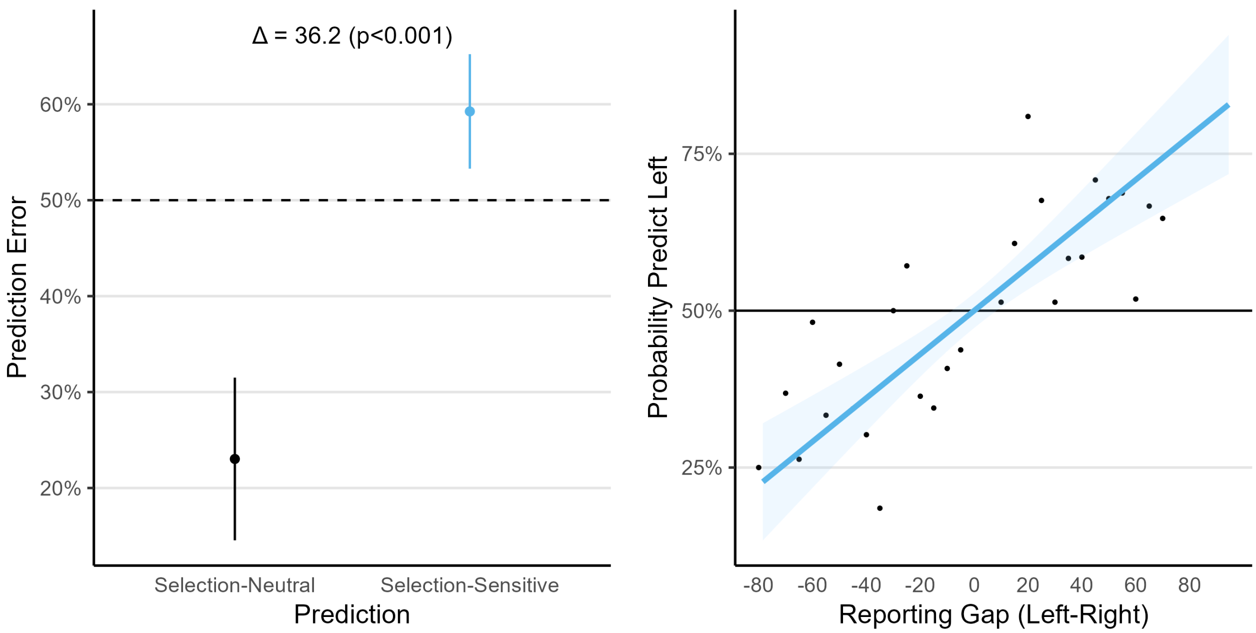
IV Selection Neglect and Biased Predictions

A Officers Neglect Selection Bias in Crime Predictions

We first address the question of whether officers neglect selection bias when making crime predictions. To do so, we focus on the rate of prediction error separately for our two types of predictions. Panel (a) of Figure 4 shows that officers make a mistake in less than a quarter of selection-neutral predictions, where adjusting for selection is not needed to give a correct answer. In contrast, error rises to 59% in selection-sensitive predictions, where the correct answer requires adjusting for selection. Thus, prediction error is 36 percentage points higher when adjusting for selection is needed. Table A.3 regresses errors on whether the predictions were selection-neutral or selection-sensitive, and confirms that excess prediction error is significant ($p < 0.001$) and robust throughout the experiment. Because the only thing that differentiates selection-sensitive and selection-neutral predictions is the need to adjust for selection, we can attribute excess prediction error to selection neglect.

Figure 4: Evidence of Selection Neglect

(a) Prediction Error by Type of Prediction **(b)** Neglect Turns Selection Bias into Prediction Bias



Notes: Panel (a) shows the average prediction error by prediction type. Selection-neutral predictions, where adjusting for selection does not change the optimal prediction, are depicted in black and in the left column; selection-sensitive predictions, where adjusting for selection does change the optimal prediction, are depicted in blue and in the right column. Error bars represent the 95% confidence interval of the mean, with standard errors clustered at the officer level. The annotation shows the difference across types of predictions, along with the p-value from a t-test clustered at the officer level (see Column (4) of Table A.3). The dashed line shows the random benchmark. Panel (b) plots the probability of predicting higher crime in the area on the left by the difference in reporting rates between areas. Points show the empirical average prediction probability at each reporting gap. The blue line shows the fitted linear regression between the two variables, holding crime fixed (see Column (5) of Table A.4). The shaded area shows the 95% confidence interval of the estimate, with standard errors clustered at the officer level. The solid horizontal line shows the optimal prediction: holding crime fixed, reporting rates are uninformative. Observations for both panels are officer-prediction pairs (N = 155 officers $\times$ 6 predictions = 930), from the first six predictions of Part II.

Selection neglect is not driven by a handful of officers: it is a widespread error. Figure A.4 shows the distribution of errors at the officer level, divided by selection-neutral and sensitive decisions. While the modal officer makes no mistakes when adjusting for selection isn't needed, they fail *all* predictions that are selection-sensitive. Focusing on the median officer reveals a similar picture: no mistakes in selection-neutral predictions, but wrong predictions in two thirds of sensitive ones. Consistently, estimates that include the 27 participants who failed the sanity check preserve the direction and statistical significance of these results but are smaller in magnitude. In the full sample, prediction error is 27.9 percentage points higher in selection-sensitive decisions. The attenuation is due to participants who failed the initial check not engaging with the task: their error rate is 60% even in selection-neutral decisions.

Neglect does not only reduce accuracy: it biases predictions toward wherever crime is more observable. Crime data shows where crime is more observable, not where it is more common, and an officer who takes reports at face value mistakes the one for the other. Table A.4 tests this by regressing officers' predictions on the reporting-rate gap between the two areas, holding the true crime gap fixed. Panel (b) of Figure 4 plots the probability of predicting higher crime in the area presented on the left as a function of the gap in reporting rates between areas, along with the estimated fit. Officers are significantly more likely to predict higher crime in the area with the higher reporting rate, conditional on the true crime difference between areas. In particular, officers become 3 percentage points more likely to predict higher crime in an area when its reporting rate becomes 10 percentage points larger. Table A.4 shows this pattern persists across both parts of the experiment and is robust to officer fixed effects. To put this magnitude in context, the difference in reporting rates between Barrios Unidos and Tunjuelito alone would make an officer 8 percentage points more likely to predict higher crime in Barrios Unidos. Note that our estimates are likely a lower bound of this cognitive error, as officers are provided all the information needed to adjust for selection bias. In the field, where this information is not that accessible, the magnitude of this error might be larger. Overall, selection neglect turns differences in crime observability into prediction bias.

So far we have shown that neglect biases crime predictions. But this error might be innocuous if it only occurred in close calls, where the two areas differ little in true crime. Figure A.5 tests this by disaggregating prediction errors by the crime gap between areas. The opposite holds: the excess prediction error caused by neglect grows with the crime gap. While selection-neutral predictions become more accurate as the gap widens, selection-sensitive predictions remain stably wrong. Weighting each incorrect prediction by the true crime gap between areas, the implied loss averages more than eight crimes per decision, over half the average crime gap in the experiment. Neglect therefore misleads officers most in precisely the decisions where being wrong is most costly.

Robustness

Are our results driven by selective effort? — One might be concerned that prediction errors are not driven by selection neglect *per se*, but by officers avoiding a costly calculation in selection-sensitive decisions. We find no evidence to support this. First, the only way to distinguish between neutral and sensitive decisions is computing the selection adjustment, as these decisions are visually identical ex ante. Second, officers spend the same time in both types of prediction ($p = 0.438$, Table A.10). Third, we also find a robust excess prediction error when restricting the sample to those who spent more time on selection-sensitive decisions. Therefore, our results do not seem to be the result of selective effort.

Are our results driven by officers taking reporting rates as a signal of crime? — If officers take higher reporting rates as a signal of crime, because they think the two are positively correlated in our experimental setting, they might predict higher crime where reporting rates are higher. The instructions already weigh against this explanation, as they state that crime and reporting rates are drawn independently in our experiment. Additionally, the relationship between reporting rates and crime in the field is usually negative, as lower observability reduces the costs for criminals (Gonçalves et al., 2026). However unlikely, Table A.5 tests this concern by regressing the predicted area on the gap in crime reports and reporting rates between areas. Columns (4) and (5) report that, conditional on crime reports, the coefficient on reporting gap is, if anything, slightly negative. In other words, officers' behavior is not consistent with them taking high reporting rates as a signal of crime.

Are our results driven by limited arithmetic skills? — A priori, this is an unlikely explanation, as the only calculation needed is a division and participating officers can use a calculator on screen. We test whether neglect is a result of poor arithmetic skills by repeating the main specification —regressing prediction error on whether the decision is selection-sensitive— and including an interaction with whether the officer correctly answered all arithmetic questions. Column (4) of Table A.6 finds significant excess prediction error among officers who correctly answer all arithmetic questions. Although the officers who make a mistake in these questions show higher excess prediction error, this difference is not statistically significant ($p = 0.166$). In other words, while arithmetic skills might help to adjust for selection, they do not explain neglect among officers.

Are our results driven by confusion about the task? — The selection neglect we identify is not explained by confusion about the task. We explained the task in detail, included several practice rounds, and officers completed nine comprehension questions. A third of officers got all of them right, while almost everyone else made one or two mistakes —which were immediately corrected. To test whether confusion plays a role in our results, Column (5) of Table A.6 regresses prediction error on whether the decision is selection-sensitive, but now adding an interaction with whether the officer made a mistake in any comprehension question. We find that selection neglect is robust among officers who answered all comprehension questions correctly, while smaller in magnitude (23 percentage points, $p = 0.030$). Again, we find higher excess prediction error among those who made some mistake in the instructions, but this difference is not significant ($p = 0.114$). Altogether, confusion about the task by itself cannot explain our results.

B Neglect and Disparities in Crime Predictions

We have used random variation in crime observability to show that area-level differences in data selection translate directly into prediction error, as officers mistake observed crime for true crime. In practice, selection does not vary randomly: there are systematic differences in observability across groups. In Colombia, crimes are more likely to be reported when they target a younger, more educated, and higher-SES victim. In the United States, Black and Hispanic individuals are overrepresented in crime data, as they are more likely to be patrolled, stopped, searched, reported in a field interview, and arrested (Chen et al., 2023; Gonçalves et al., 2025; National Academies of Sciences, Engineering, and Medicine, 2023). If officers neglect that crime data oversamples one group, officers will predict higher crime among the group where crime is more observable. In other words, neglect can turn differences in crime observability into disparities in predictions.

We directly test this implication by inducing systematic differences in observability across *cities* in our experiment. Areas within the same city, which acts as an abstract group, share the same distribution of crime; participants are aware of this. We implement two conditions. In the baseline *No Selection Bias Across Groups*, observability of crime is identical across cities: reporting rates are drawn from the same distribution. In contrast, in the *Selection Bias Across Groups* treatment, reporting rates in one city are 30 percentage points higher on average[6]. Importantly, underlying crime distributions remain identical across cities. Thus, in this treatment cities systematically differ in the number of crimes reported, but this difference is entirely driven by the gap in observability. Under this design, systematic prediction bias towards one city is an inference error that can be solely attributed to selection neglect. We randomize officers either into the Baseline or the Selection Bias Across Groups condition for all twelve decisions of Part II. In this section, we focus on the first six of these predictions, as the last six also include the algorithmic recommendation treatments.

Do officers adjust for selection bias across groups, or do they carry it into biased beliefs? Because crime is identical across cities, a rational decision maker should be equally likely to predict higher crime in either city. In contrast, a neglecting decision maker would confound observability with crime, and thus predict higher crime where it is more likely to be reported. To test this prediction, we classify, for each participant, which city had a higher reporting rate on average [7]. Figure 5 plots how likely officers are to predict higher crime in each city, divided by panels according to whether there is systematic selection bias across cities. When crime is equally observable–same average reporting rate–in both cities (left panel), officers are equally likely to predict higher crime in either ($p = 0.304$). In contrast, when crime is more observable in one city, officers are 20 percentage points more likely to predict more crime in the group where it is more observable ($p < 0.001$). In other words, officers mistake having a higher number of crime reports among one group for crime being higher among that group, even when both groups have identical crime levels.

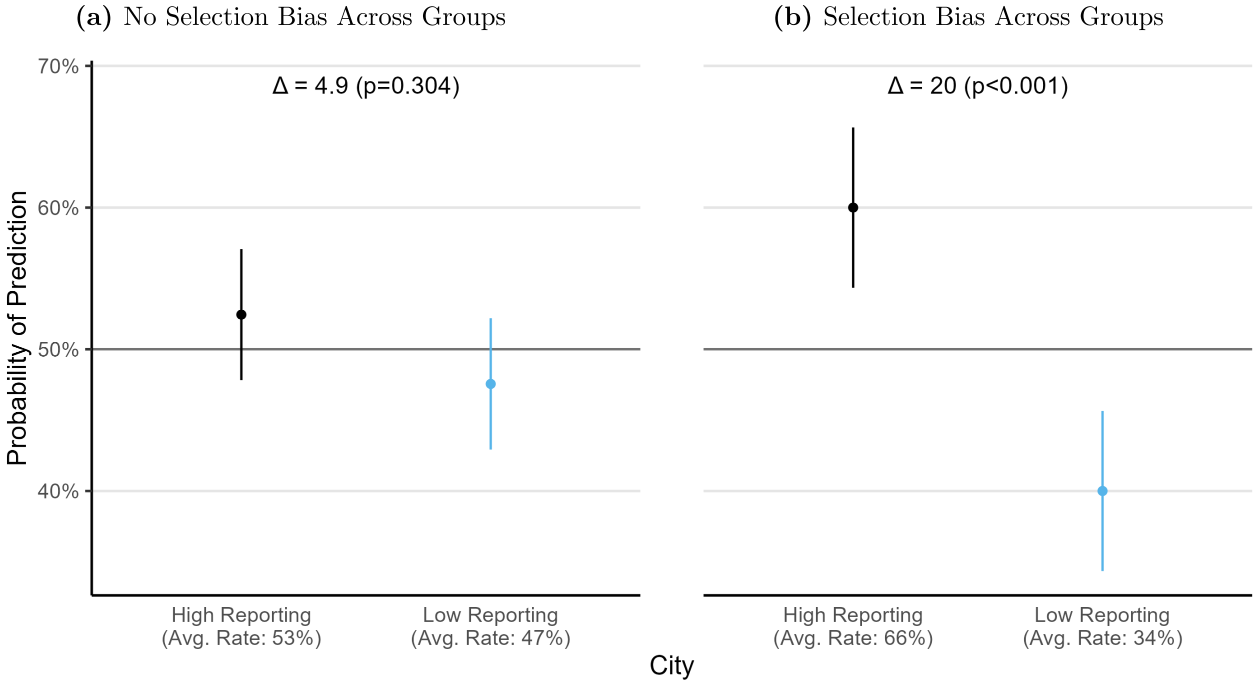
This prediction bias error is not an artifact of the empirical realization of crimes, nor is it driven by a handful of officers. One concern is that, despite the distribution of crime being identical across cities, the empirical realization of these distributions makes the city with higher reporting rates also the city with more crime across these six predictions. However, the opposite holds: in both conditions, the city with lower reporting rates happens to have *more* crime on average. When we benchmark the prediction bias across groups against the empirically optimal gap, the bias towards the city where crime is more observable only increases (see Table A.7). Figure A.6 plots the distribution of prediction bias at the officer-level. Both measures of bias —the raw prediction gap between cities and the gap adjusted by crime realizations— show a similar picture: the majority of officers predict more crime in the group where crime is more observable.

Comparing predictions with the empirical optimum reveals that the prediction bias caused by neglect

[6] In particular, reporting rates for both cities are uniformly distributed between 10% and 90% in the baseline, and rounded to the closest multiple of five to ease calculations. In the Selection Bias Across Groups condition, reporting rates for one city are uniformly drawn between 10% and 60%; between 40% and 90% for the other, again rounded to multiples of five. Which city is chosen as the high-reporting one is a coin flip. For more details, see Appendix B.

[7] At baseline, both cities have the same reporting rate in expectation. However, because of randomization, one city always has a slightly higher reporting rate on average. We take this city as the city with higher average reporting rates in the No Selection Bias Across Groups condition.

Figure 5: Prediction bias across groups



Notes: This figure shows how likely the average officer is to predict higher crime in each of the two cities. Cities act as abstract groups to which areas belong. We classify each of the two cities, separately for each participant, as High or Low Reporting, according to which had the higher average reporting rate across the first six predictions of Part II. The figure is divided in panels according to whether both cities have the same reporting rates in expectation (No Selection Bias Across Groups condition), or one city had 30 percentage points higher reporting rates in expectation (Selection Bias Across Groups condition). Error bars show the 95% confidence interval of the mean, with standard errors clustered by officer. The annotations show the difference in prediction probability across cities, along with the p-value of a t-test clustered at the officer level (see Table A.7 for details). Observations are officer-prediction pairs (left panel N = 75 officers × 6 predictions = 450; right panel N = 80 officers × 6 decisions = 480), from the first six predictions of Part II.

is inference error, i.e., *inaccurate*. There is no true difference in crime that a higher prediction for the more-observable group could reflect. This distinction matters for how we interpret discrimination. Tests for discrimination often pit taste-based discrimination against accurate statistical discrimination: if an officer's behavior cannot be explained by rational expectations, it is attributed to animus (Anwar & Fang, 2006; Knowles et al., 2001). Our results show a third possibility. Officers in this task hold no animus, as groups are abstract, and yet their predictions are inaccurately biased against one group. These belief disparities provide a cognitive foundation for inaccurate statistical discrimination. When crime predictions guide enforcement decisions, selection neglect can translate differences in observability into differential treatment even without prejudice or accurate group differences.

This disparity in crime predictions requires strikingly little. There are no stereotypes, no group-specific preferences, and no true difference in crime across groups. A single inference error, taking selected data at face value, is enough. In the field, where observability differs across real groups — socioeconomic status or race — the same mechanism would push predictions against whichever group is more heavily represented in the data. We return to what this implies for policing disparities in Section VI.

V Mechanisms

A Framework

We now delineate a simple conceptual framework to build intuition about the possible mechanisms of selection neglect. In our task, making the correct crime prediction requires two steps. First, an officer must correctly *represent* the inference problem, that is, recognize that crime data is a selected sample that is not representative of underlying crime. Second, conditional on this representation, the officer must *compute* correctly the adjustment for selection. Neglect can thus arise from a failure at either of these steps, representation or computation.

A key marker of a failure of representation is information demand. If an officer does not incorporate selection bias into their inference process, they will not demand this information when making crime predictions. We directly elicit this demand by letting officers choose whether to see reporting rates or not in one crime prediction. There are two non-exclusive explanations for why officers represent the problem without considering selection bias: bottom-up and top-down inattention. Salience might drive bottom-up attention to other features of crime data at the expense of selection (Bordalo et al., 2026). On the other hand, officers might hold mental models derived from their experience with this inference problem, which they routinely tackle in the field (Schwartzstein, 2014). If their mental model of crime data does not include its selection, they will not be attentive to this information. We test both sources of inattention in our experiment.

Neglect can also arise at the computation step, either while making the calculation itself or from officers anticipating the complexity of the adjustment and thus opting to apply a simpler heuristic (Arrieta & Nielsen, 2024). While these explanations are unlikely in the Crime Prediction Task, which only requires a simple division, we provide a battery of evidence against neglect being driven by failures at the computation step.

B Neglect as a representational failure

We begin by examining the first mechanism: representation of the inference problem. A first marker of this failure is information demand: an officer who chooses not to see reporting rates —the information needed to adjust for selection bias— is not using it as input in their inference. We test this implication directly, letting officers choose whether to see reporting rates for the last decision of Part I. Figure A.7 shows that 36% of officers choose not to see reporting rates for this decision. Moreover, this failure not only arises in this last prediction. Table A.5 regresses officers' predictions on the difference between areas for every piece of information. We find that officers follow the difference in crime reports to guide their predictions, but, conditional on this difference, they do not change their predictions as much by the reporting gap between areas.[8] This is especially the case among officers who choose not to see reporting rates in the last prediction of this part, confirming that information demand corresponds to information

[8]The expected sign of the reporting-rate gap depends on the conditioning set. Conditional on the difference in reports, an officer who adjusts for selection should be *less* likely to predict the area with the higher reporting rate. Adjusting for selection means inferring Crimes = Reports/Reporting Rate: holding reports fixed, a higher reporting rate implies fewer underlying crimes, as the same number of reports now represents a larger share of all crimes. A coefficient of zero thus indicates that reporting rates do not affect officers' inference, while a positive coefficient would indicate that officers read high reporting rates as a signal of crime.

use across the entire experiment. In other words, many officers do not seem to use reporting rates when making crime predictions, and over a third of them do not even demand this information.

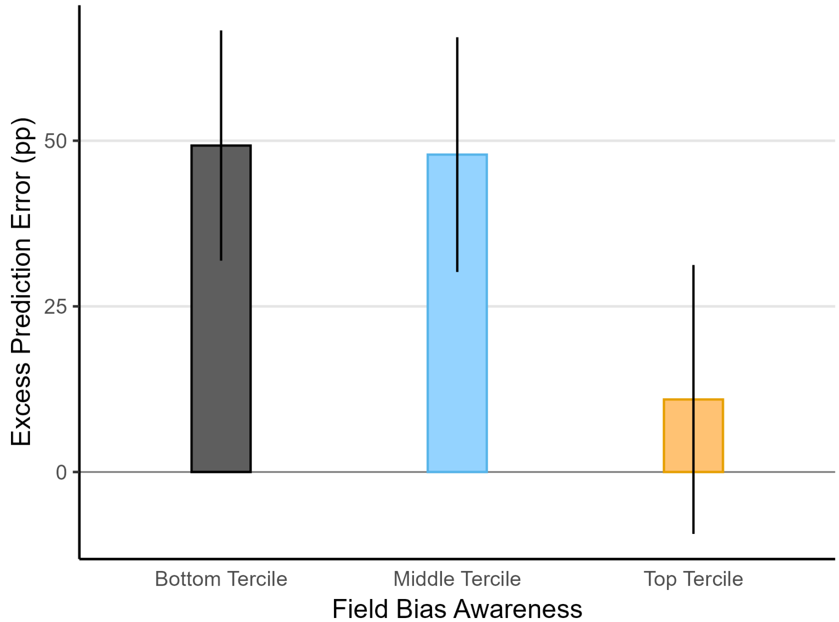
Importantly, past experience with crime predictions *in the field* is correlated with information demand. Table A.8 regresses the choice to view reporting rates on instruction comprehension, arithmetic skills, professional experience, and awareness of bias in crime data. We find that ignoring selection is not explained by confusion about the task or lack of basic numeracy. Instead, it varies with officers' previous assignment. Officers whose most recent assignments involved surveillance and patrol assignment (tasks that rely heavily on predicting crime from reported crime data) are 20 percentage points ($p = 0.035$) more likely to view reporting rates than administrative officers. This correlational evidence points to officers forming their mental model about crime prediction and crime data from their experience in the field.

What explains the failure to represent the inference problem correctly? One potential mechanism is salience. In the field, officers might ignore reporting rates because these are not included in official statistical systems, and thus focus instead on more accessible information. While this cannot explain our main results, as we make reporting rates easily available throughout the experiment, we now explore whether their salience is related to neglect. In Part I, reporting rates and crime reports are shown along multiple uninformative variables; in Part II, these decoys are removed, making selection visually unavoidable. Increasing the salience of reporting rates does not make officers incorporate them in their inference process. Figure A.8 plots the excess prediction error caused by neglect in both parts of the experiment, finding similar magnitudes for both salience levels. Column (6) of Table A.5 finds that participants who chose not to see reporting rates, who are a priori the most likely beneficiaries of salience, still fail to use information about selection when made more salient. Overall, the lack of salience of reporting rates does not seem to explain neglect: officers neglect the selective observability of crime even when made more observable.

At the end of the experiment, we ask officers their opinion on five statements about bias in reported crime data, the data they routinely work with. These statements include whether crime reports accurately reflect the (i) spatial and (ii) temporal trends of crime; (iii) whether citizens know how to report a crime; (iv) whether reported crime data is the best source to predict crime, and (v) whether it provides a biased map of underlying crime. These five questions tap an underlying belief that crime reports misrepresent true crime, and because the answers are highly correlated, we aggregate them into a single index following Kling et al. (2007), reducing measurement error from any single item. We denote this index *field bias awareness*, where higher values reflect greater recognition that reported crime data is a biased sample of crime.

We use the *field bias awareness* index to further test whether officers are applying an incorrect mental model they bring from their work in the field. We divide the sample into terciles of the index and estimate the excess prediction error caused by selection separately for each group. Figure 6 shows that among the third of officers with the highest awareness, neglect increases prediction error by only 10 percentage points ($p = 0.298$). In contrast, excess prediction error reaches nearly 49 percentage points among the third of the sample with the lowest awareness. In other words, officers who are more aware that reported crime data *in the field* is a biased dataset are able to adjust for selection bias *in the lab*, when given the information to do so. This correlation supports the interpretation that neglect arises from the wrong mental model of crime data and, together with the previous result on past assignment, suggests these models can develop

Figure 6: Selection Neglect by Field Bias Awareness



Notes: The figure shows the excess prediction error caused by selection neglect, estimated separately by terciles of Field Bias Awareness. In particular, we regress prediction error on whether the prediction is selection-sensitive, interacted with a factor variable that divides the sample in terciles. The bars represent the coefficient of selection-sensitivity, along with error bars that show its 95% confidence interval with standard errors clustered at the officer level. Observations are officer-prediction pairs (N = 153 officers × 6 predictions = 918; excludes two officers who did not answer the Field Bias Awareness questions), from the first six predictions of Part II. See Columns (5) and (6) of Table A.9 for similar regressions using a continuous measure of Field Bias Awareness.

from experience in the field.

A final piece of evidence comes from the arithmetic questions. We included in the battery a question that closely resembles the Crime Prediction Task, yet with a different frame from crime. In particular, we ask officers "We counted the male students in a class, so we only counted two thirds of the class. If there were 40 men in that class, how many people took that class in total?". This question captures the exact adjustment needed in the Crime Prediction Task. When asked in a neutral way, over 75% of officers are able to recover the whole from an observed subsample and get this question right. However, these same officers fail to apply the same adjustment in the Crime Prediction Task. Column (6) of Table A.6 regresses prediction errors on whether the prediction was selection-sensitive but focusing only on officers who get the neutral question right, and finds an excess prediction error of 32 percentage points ($p < 0.001$). Along with the previous results, this suggests that officers are able to perform the necessary computations, but fail to represent the problem correctly when framed as a crime prediction. We interpret this result as officers applying their (wrong) mental models of crime data to the prediction task when framed as a crime prediction.

In general, we find little evidence to support the role of computational frictions behind neglect. First, an algorithm that completely eliminates calculation costs does not improve crime predictions nor reduce neglect, as explained in detail in the following subsection. Second, the computation needed is a simple division, and participants have a calculator on screen. However, only 30% of officers ever use it during the main task, and this choice is positively correlated with field bias awareness ($p = 0.024$). Thus, although Table A.3 finds no neglect among participants who used the calculator, we cannot rule out that this better performance is driven by officers who correctly represent the problem selecting into using the calculator. Overall, the evidence suggests that officers who represent the problem correctly use the calculator to reduce frictions, but these frictions alone cannot account for the excess prediction error caused by selection.

Selection Neglect and Advice Override

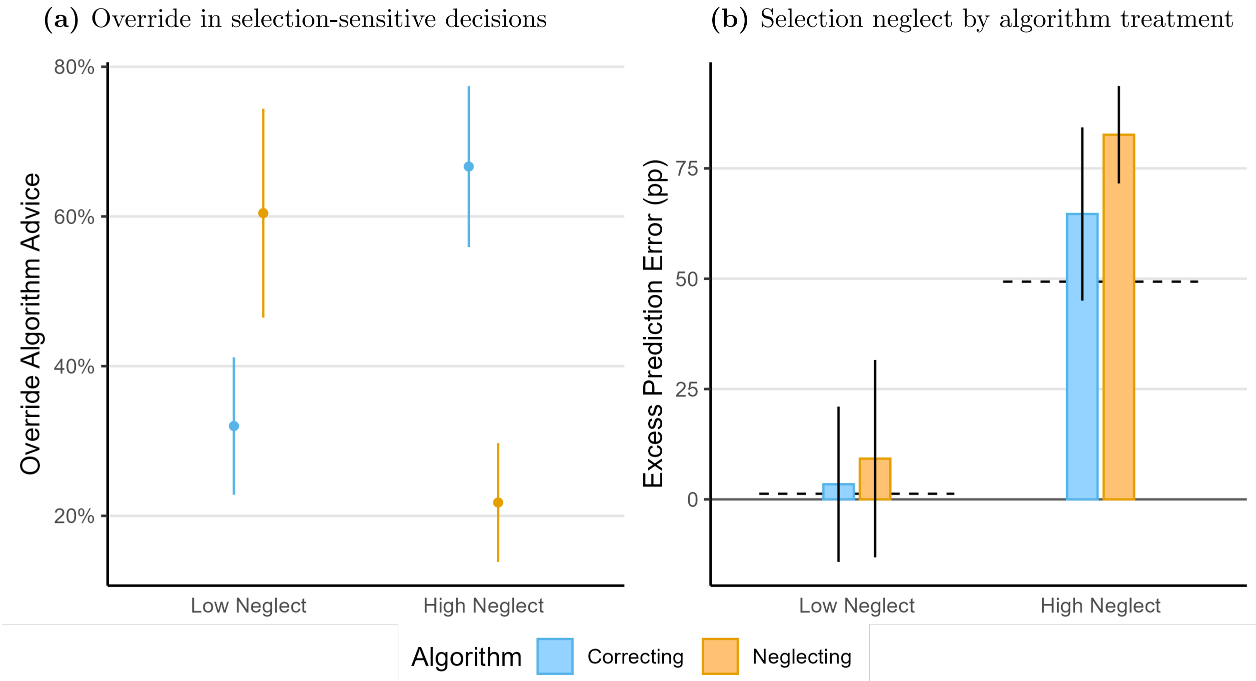
We next introduce algorithmic recommendations as a test of the mechanism underlying selection neglect. For the last six predictions of Part II, officers are randomly assigned to one of three conditions: a control group with no recommendation, a correcting algorithm that adjusts reported crime for reporting rates, and a neglecting algorithm that takes reports at face value. Officers are informed if and how their algorithm adjusts for reporting rates and they retain full discretion to follow or override its recommendations. Because officers retain discretion, adherence and override patterns reveal whether algorithms substitute for inference or are filtered through existing mental models.

We begin by dividing the sample based on their predictions before the algorithm conditions. Using the first six decisions in Part II, before algorithms are introduced, we classify officers into two groups: those with above the median prediction error in selection-sensitive decisions (high neglect), and those below the median (low neglect). This classification broadly captures whether the inference process of officers was correct before receiving any algorithmic advice.

We first analyze whether officers follow or override recommendations. Panel (a) of Figure 7 reports the fraction of algorithmic recommendations that officers choose to override, focusing on selection-sensitive predictions —where each algorithm gives opposite advice. Officers who show low neglect are twice as likely to override the advice of the neglecting algorithm than the correcting one. In contrast, officers with high neglect display the opposite pattern: they are more than three times as likely to override the correcting algorithm than the neglecting one. In other words, officers tend to ignore the advice that conflicts with their prior behavior. Moreover, Figure A.9 shows that higher field bias awareness is associated with more override of the neglecting algorithm and more adherence to the correcting one. This correlation further supports the interpretation that officers override advice that conflicts with their *representation* of the crime prediction problem.

These adherence and override patterns have clear implications for the effects of the algorithms on selection neglect. Panel (b) of Figure 7 reports excess prediction error by algorithm condition and the prior level of neglect. Among officers with low prior neglect, excess prediction errors remain low across all conditions, as officers reject the advice of the algorithm that takes reports at face value. However, officers with high prior neglect override the corrective advice, slightly increasing excess prediction errors with respect to the Control group ($p = 0.264$). In other words, officers fail to adjust for selection bias in crime data even when given the correct answer. These same officers overwhelmingly follow the advice of the Neglecting algorithm, which takes reports at face value, increasing their excess prediction errors by over 30 percentage points ($p = 0.003$). Because officers only follow advice aligned with their preconceived mental model, algorithmic advice fails to correct neglect. Instead, algorithmic advice that does not adjust

Figure 7: Override of algorithm advice and selection neglect



Notes: Panel (a) shows the share of selection-sensitive decisions in which officers override the algorithm recommendation, dividing the sample in half according to their level of neglect in the main analysis. The left column shows officers with Low Neglect —below-median prediction error ($< 66\%$) in selection-sensitive decisions, while the right column shows officers with High Neglect —above-median prediction error. Error bars show the 95% confidence interval of the mean, with standard errors clustered at the officer level. Panel (b) shows the excess prediction error due to selection —the estimated coefficient of selection-sensitivity on prediction error, disaggregated by algorithm intervention and level of neglect. The dashed lines represent the excess prediction error of officers in each group that were randomized to the control algorithm (no algorithm advice). Error bars show the 95% confidence interval of the coefficient, with standard errors clustered at the officer level. Colors show the algorithm treatments: Control in black dashed lines, Correcting in blue, and Neglecting in orange. Observations in both panels are officer-prediction pairs (N = 155 officers $\times$ 6 predictions = 930; with 55 officers assigned to Control, 48 to Correcting, and 52 to Neglecting), from the last six predictions of Part II. The left panel subsets this sample to the 370 predictions that are selection-sensitive, excluding officers in the Control condition.

for selection bias can amplify neglect by validating the wrong model.

The evidence from offering officers algorithmic advice supports the interpretation of neglect as a representational failure. If neglect was arising from computational frictions, officers should follow the correcting algorithm, eliminating excess prediction errors. Instead, algorithmic advice is filtered through existing mental representations. Information provision that does not change the underlying representation of how crime data are generated is therefore unlikely to eliminate selection neglect, and may even reinforce it.

Four pieces of evidence converge on the same conclusion: selection neglect is a failure of representation, not of information, attention, or computation. First, officers do not demand the information needed to adjust for selection: over a third choose not to see reporting rates, and those who do not look also do not use them when available. Selection is just not part of how they represent the inference problem. Second, this representation tracks officers' experience with crime data in the field. Officers whose assignments involve predicting crime from reports are more likely to consult reporting rates, and officers who recognize the bias of crime data in the field are better at adjusting for selection bias in the lab. The mental model that officers bring from the field predicts how officers use the crime data presented to them. Third, officers who fail to adjust are not unable to compute the adjustment. They correctly perform the same adjustment when it is posed as a neutral arithmetic problem, and they largely ignore an on-screen calculator that would eliminate any computational cost. What they lack is not the capacity to compute the adjustment, but a representation of the inference problem that calls for it. Fourth, the algorithm conditions confirm the representational nature of neglect. If neglect were computational, officers would welcome an algorithm that performs the adjustment for them; instead, high-neglect officers override the correcting algorithm and follow the neglecting one, adopting advice only when it matches the model they already hold. This is the sense in which selection neglect is representational: it is not that officers cannot compute, but that selection does not enter their model of what crime data captures about underlying crime.

VI Discussion

Police departments increasingly rely on crime data to make crime predictions and guide enforcement decisions (Brayne, 2020; Perry et al., 2013). However, crime data often suffers from selection bias: not all crimes are observed, and observability depends on the victim, the perpetrator, the type of crime, and where it was committed. This paper shows that senior police officers do not account for selection bias when inferring crime, turning selection bias into biased predictions. We now discuss the broader implications of our findings.

In Colombia, selection bias in crime data comes from selective reporting behavior. The good news is that this bias is measurable through victimization surveys. Addressing the issue of selected data thus requires better measurement of the patterns of selection bias: more frequent and finer-grained victimization surveys that allow a better estimation of reporting rates. At the same time, these surveys offer an alternative metric of crime —victimization—, which can complement the use of selection-adjusted crime reports. However, our results provide a cautionary tale: mental models about crime data might be a binding constraint to data-driven policing. We find that simply providing the necessary information for officers to adjust, or even the advice of a perfectly predictive algorithm, does not lead to better predictions, as officers evaluate the new information through mental models that are correlated with their prior field experience. Our findings suggest that improving the data ecosystem may therefore need to be accompanied by interventions that change how officers interpret selected data.

Providing better data alone may therefore be insufficient to reduce selection neglect. The potential consequences of neglect for effective and equitable policing are clear. We show that neglect is *sufficient* to generate biased predictions, absent animus, stereotypes, or true crime differences. We do not address whether this mechanism explains observed disparities in policing, nor directly measure its magnitude in the field, because of the endogeneity of crime and observability in administrative data. However, note that the extent of neglect identified in our experiment is likely to be a lower bound. We create the most favorable conditions for adjusting for selection bias: officers learn the data generating process, have precise information about observability in each area, are incentivized for accuracy, and have no institutional constraints. Despite the favorable conditions, neglect still biases crime predictions. In the field, where none of these conditions hold, neglect is likely to be more pervasive.

The mechanism we identify in this paper is not specific to Colombia or to selection driven by reporting behavior. Whenever crime observability is incomplete and biased across groups, selection neglect can translate those differences into biased predictions. Enforcement-driven selection is an especially pernicious case of this mechanism. Police departments across the United States often use police-generated data to make crime predictions, such as police reports or field interviews (Brayne, 2020; Perry et al., 2013). Because officers disproportionately patrol and stop minority individuals (Chen et al., 2023; Pierson et al., 2020), minorities become overrepresented in crime data, which then justifies further enforcement through higher predicted crime. When selection is itself generated by enforcement, the form of selection neglect documented here can contribute to the feedback loops described by Che et al. (2026) and Hübert and Little (2023). Overall, our findings identify selection neglect as a cognitive mechanism that can sustain policing disparities through inaccurate statistical discrimination.

Our results not only identify a cognitive mechanism that can feed policing disparities, they also question the methods to identify their source. Traditional outcome tests of bias in policing assume that officers make rational predictions from observable data (Anwar & Fang, 2006; Becker, 1971; Knowles et al., 2001). If officers' behavior cannot be explained by accurate crime predictions across racial groups, for instance, by accurately anticipating higher criminality among one group, their behavior is attributed to racial animus. However, there is a third possibility: officers might make *inaccurate* crime predictions because of cognitive failures (Bohren et al., 2023). Omitting this possibility can lead analysts to mistake wrong inference for racial animus. By eliciting crime predictions directly from senior police officers, this paper provides direct evidence that officers make *inaccurately* biased predictions about crime. Our results are consistent with Arnold et al. (2018), whose results suggest inaccurate stereotypes drive racial bias in bail decisions. More broadly, our paper contributes to a growing literature that seriously addresses the assumptions of traditional tests, like the selection bias in administrative data (Durlauf & Heckman, 2020; Knox et al., 2020), the ability of officers to accurately target their decisions according to potential outcomes (Feigenberg & Miller, 2022; Rose, 2021), or the actual link between potential outcomes and decisions (Canay et al., 2024; Stashko, 2023). Going forward, analyses of the sources of policing disparities need to take seriously the fact that officers may hold inaccurate beliefs.

That officers do not draw accurate inferences from selected data should not be surprising. Adjusting for selection bias in administrative crime data is an issue even for highly trained researchers (Gonçalves et al., 2025) and unbiased predictive algorithms (Arnold et al., 2025), and many lab experiments document the pervasiveness of this failure in more abstract settings (Ali et al., 2021; Araujo et al., 2021; Barron et al., 2024; Enke, 2020; Esponda & Vespa, 2018; Farina & Herman, 2025; Jin et al., 2021). We extend these findings from the lab and identify selection neglect among expert practitioners, who routinely make predictions from selected data in the field. Expert practitioners bring mental models developed with experience that lab participants lack. Our results show that these mental models might not only be the source of cognitive failures, but also a constraint for interventions that address them. Providing officers with the exact data generating process, making selection more salient, or even giving them the correct answer fail to improve predictions, consistent with these interventions being filtered through prior mental models. In contrast, similar interventions have been found to work to debias experimental subjects, who lack these field-developed mental models. For instance, Enke (2020) finds that making selection bias more salient curbs neglect, and López-Pérez et al. (2022) find that more transparent and simpler selection processes reduce neglect. Along these lines, our results warn about the limits of debiasing methods developed in the lab, as they might not transfer to professionals with entrenched representations.

Crime data reflects where crimes are observed, not where they happen. This paper shows that officers take selected crime data at face value, turning selected data into biased predictions. This mechanism requires no prejudice or stereotypes, yet it is enough to bias officers' predictions and can potentially mislead tests that attribute such bias to its source. You do not need a biased cop to get biased beliefs: bad inference is enough.

References

Ali, S. N., Mihm, M., Siga, L., & Tergiman, C. (2021). Adverse and Advantageous Selection in the Laboratory. *American Economic Review*, *111*(7), 2152–2178.

Ang, D., Bencsik, P., Bruhn, J., & Derenoncourt, E. (2025). Community Engagement with Law Enforcement after High-Profile Acts of Police Violence. *American Economic Review: Insights*, *7*(1), 124–142.

Angelova, V., Dobbie, W., & Yang, C. S. (2025). Algorithmic Recommendations and Human Discretion. *The Review of Economic Studies*, rdaf084.

Anwar, S., & Fang, H. (2006). An Alternative Test of Racial Prejudice in Motor Vehicle Searches: Theory and Evidence. *American Economic Review*, *96*(1), 127–151.

Araujo, F. A., Wang, S. W., & Wilson, A. J. (2021). The Times They Are A-Changing: Experimenting with Dynamic Adverse Selection. *American Economic Journal: Microeconomics*, *13*(4), 1–22.

Arnold, D., Dobbie, W., & Hull, P. (2025). Building Nondiscriminatory Algorithms in Selected Data. *American Economic Review: Insights*, *7*(2), 231–249.

Arnold, D., Dobbie, W., & Yang, C. S. (2018). Racial Bias in Bail Decisions. *The Quarterly Journal of Economics*, *133*(4), 1885–1932.

Arrieta, G., & Nielsen, K. (2024). Procedural Decision-Making In The Face Of Complexity.

Barron, K., Huck, S., & Jehiel, P. (2024). Everyday Econometricians: Selection Neglect and Overoptimism When Learning from Others. *American Economic Journal: Microeconomics*, *16*(3), 162–198.

Becker, G. S. (1971, August). *The Economics of Discrimination* (2. edition, Ed.). University of Chicago Press.

Bhuller, M., & Sigstad, H. (2024). Feedback and Learning: The Causal Effects of Reversals on Judicial Decision-Making. *The Review of Economic Studies*, rdae073.

Bohren, J. A., Haggag, K., Imas, A., & Pope, D. G. (2023). Inaccurate Statistical Discrimination: An Identification Problem. *The Review of Economics and Statistics*, 1–45.

Bordalo, P., Conlon, J., Gennaioli, N., Kwon, S., & Shleifer, A. (2026). How People Use Statistics. *The Review of Economic Studies*, *93*(1), 250–285.

Brayne, S. (2020, November). *Predict and Surveil: Data, Discretion, and the Future of Policing*. Oxford University Press.

Canay, I. A., Mogstad, M., & Mountjoy, J. (2024). On the Use of Outcome Tests for Detecting Bias in Decision Making. *Review of Economic Studies*, *91*(4), 2135–2167.

CEJ. (2025). *Informe de estadísticas del SPOA 2025* (tech. rep.).

Che, Y.-K., Kim, J., & Mierendorff, K. (2026, May). Data-Driven Monitoring and Deterrence in a Changing Environment.

Chen, M. K., Christensen, K. L., John, E., Owens, E., & Zhuo, Y. (2023). Smartphone Data Reveal Neighborhood-Level Racial Disparities in Police Presence. *The Review of Economics and Statistics*, 1–29.

Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, *92*(1), 1–45.

Cruces, G., Perez-Truglia, R., & Tetaz, M. (2013). Biased perceptions of income distribution and preferences for redistribution: Evidence from a survey experiment. *Journal of Public Economics*, *98*, 100–112.

DANE. (2023). *Encuesta de convivencia y seguridad ciudadana (ECSC)* (tech. rep.).

DellaVigna, S., & Kaplan, E. (2007). The Fox News Effect: Media Bias and Voting. *The Quarterly Journal of Economics*, *122*(3), 1187–1234.

Dube, O., MacArthur, S. J., & Shah, A. K. (2024). A Cognitive View of Policing. *The Quarterly Journal of Economics*.

Durlauf, S. N., & Heckman, J. J. (2020). An Empirical Analysis of Racial Differences in Police Use of Force: A Comment. *Journal of Political Economy*, *128*(10), 3998–4002.

Enke, B. (2020). What You See Is All There Is. *The Quarterly Journal of Economics*, *135*(3), 1363–1398.

EPV. (2025, November). *Encuesta de Percepción y Victimización de Bogotá 2025* (tech. rep.).

Esponda, I., & Vespa, E. (2018). Endogenous sample selection: A laboratory study. *Quantitative Economics*, *9*(1), 183–216.

Esponda, I., Vespa, E., & Yuksel, S. (2024). Mental Models and Learning: The Case of Base-Rate Neglect. *American Economic Review*, *114*(3), 752–782.

Farina, A., & Herman, C. (2025). Communicating with Data Generating Processes: An Experimental Analysis.

Feigenberg, B., & Miller, C. (2022). Would Eliminating Racial Disparities in Motor Vehicle Searches have Efficiency Costs? *The Quarterly Journal of Economics*, *137*(1), 49–113.

Ferrazares, T. (2025). Shift Structure and Cognitive Depletion: Evidence from Police Officers.

Glaser, J. (2015). *Suspect Race: Causes and Consequences of Racial Profiling*. Oxford University Press.

Gonçalves, F., Jácome, E., & Weisburst, E. (2026). Community Engagement and Public Safety: Evidence from Crime Enforcement Targeting Immigrants. *American Economic Review*, *116*(8), 3076–3109.

Gonçalves, F., Mello, S., & Weisburst, E. K. (2025, August). *Selection Bias and Racial Disparities in Police Use of Force* (tech. rep. No. w34175). National Bureau of Economic Research.

Hübert, R., & Little, A. T. (2023). A Behavioural Theory of Discrimination in Policing. *The Economic Journal*, *133*(655), 2828–2843.

Jin, G. Z., Luca, M., & Martin, D. (2021). Is no news (perceived as) bad news? An experimental investigation of information disclosure. *American Economic Journal: Microeconomics*, *13*(2), 141–173.

Kling, J. R., Liebman, J. B., & Katz, L. F. (2007). Experimental Analysis of Neighborhood Effects. *Econometrica*, *75*(1), 83–119.

Knowles, J., Persico, N., & Todd, P. (2001). Racial Bias in Motor Vehicle Searches: Theory and Evidence. *Journal of Political Economy*, *109*(1), 203–229.

Knox, D., Lowe, W., & Mummolo, J. (2020). Administrative Records Mask Racially Biased Policing. *American Political Science Review*, *114*(3), 619–637.

Koehler, J. J., & Mercer, M. (2009). Selection Neglect in Mutual Fund Advertisements. *Management Science*, *55*(7), 1107–1121.

López-Pérez, R., Pintér, Á., & Sánchez-Mangas, R. (2022). Some conditions (not) affecting selection neglect: Evidence from the lab. *Journal of Economic Behavior & Organization*, *195*, 140–157.

Lum, K., & Isaac, W. (2016). To Predict and Serve? *Significance*, *13*(5), 14–19.

Malmendier, U., & Shanthikumar, D. (2007). Are small investors naive about incentives? *Journal of Financial Economics*, *85*(2), 457–489.

National Academies of Sciences, Engineering, and Medicine. (2023, May). *Reducing Racial Inequality in Crime and Justice: Science, Practice, and Policy*. National Academies Press.

Owens, E., Weisburd, D., Amendola, K. L., & Alpert, G. P. (2018). Can You Build a Better Cop? *Criminology & Public Policy*, *17*(1), 41–87.

Perry, W. L., McInnis, B., Price, C. C., Smith, S., & Hollywood, J. S. (2013, September). *Predictive Policing: The Role of Crime Forecasting in Law Enforcement Operations* (tech. rep.).

Pierson, E., Simoiu, C., Overgoor, J., Corbett-Davies, S., Jenson, D., Shoemaker, A., Ramachandran, V., Barghouty, P., Phillips, C., Shroff, R., & Goel, S. (2020). A large-scale analysis of racial disparities in police stops across the United States. *Nature Human Behaviour*, *4*(7), 736–745.

PNVCC. (2010). *Plan Nacional de Vigilancia Comunitaria por Cuadrantes* (tech. rep.).

Rodríguez Cuadrado, J. F. (2008). La historia y el presente de las cifras delictivas y contravencionales en Colombia: Un nuevo conocimiento. *Revista Criminalidad*, *50*(1), 109–118.

Rose, E. K. (2021). Who Gets a Second Chance? Effectiveness and Equity in Supervision of Criminal Offenders. *The Quarterly Journal of Economics*, *136*(2), 1199–1253.

Schwartzstein, J. (2014). Selective Attention and Learning. *Journal of the European Economic Association*, *12*(6), 1423–1452.

Stashko, A. (2023). Do Police Maximize Arrests or Minimize Crime? Evidence from Racial Profiling in U.S. Cities. *Journal of the European Economic Association*, *21*(1), 167–214.

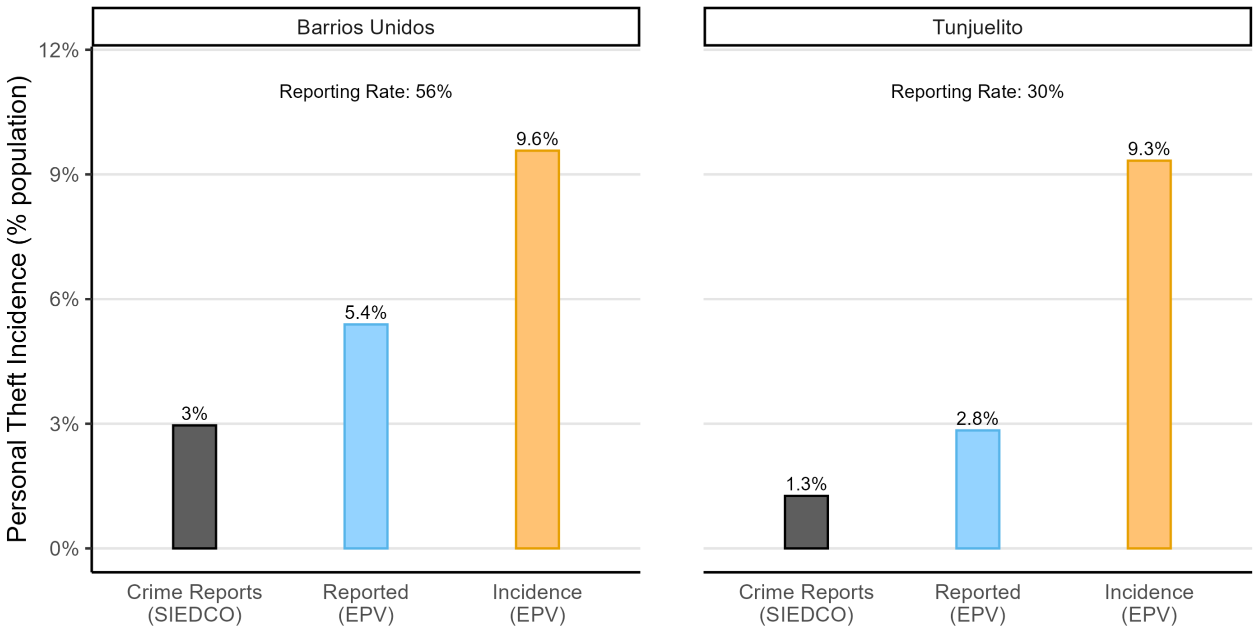
Vomfell, L., & Stewart, N. (2021). Officer bias, over-patrolling and ethnic disparities in stop and search. *Nature Human Behaviour*, *5*(5), 566–575.

Weisburd, S. (2021). Police Presence, Rapid Response Rates, and Crime Prevention. *The Review of Economics and Statistics*, *103*(2), 280–293.

A Appendix

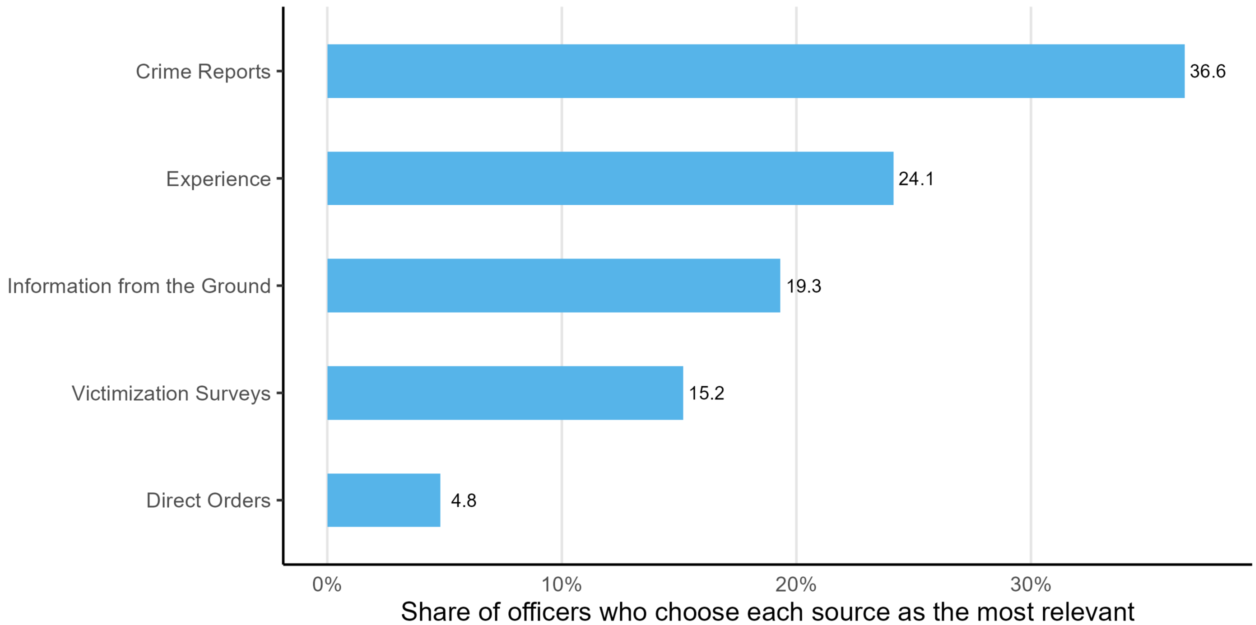
A Additional Figures and Tables

Figure A.1: Crime Data on Theft, Barrios Unidos and Tunjuelito 2025



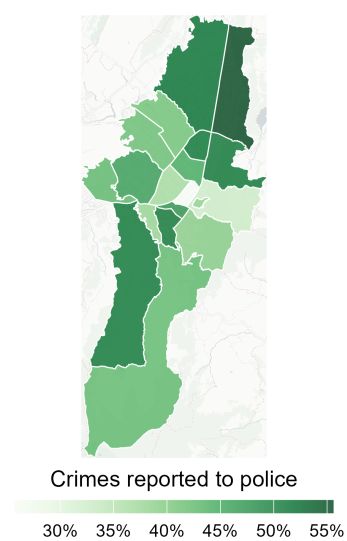
Notes: Personal theft in two Bogotá localities with near-identical survey victimization. Each panel shows three bars, all expressed as a percentage of the locality's population. Crime Reports (SIEDCO) is the count of personal thefts officially reported to the police in the locality during 2025 (SIEDCO administrative data), divided by locality population. Reported (EPV) is the share of respondents who were victimized and say they reported the crime to the police, from the 2025 Encuesta de Percepción y Victimización (EPV). Incidence (EPV) is the share of survey respondents who report having been a victim of personal theft, from the 2025 Encuesta de Percepción y Victimización (EPV); because the EPV is representative at the locality level, this share estimates the victimization rate of the locality's residents. The label above each panel is the reporting rate among residents, Reported (EPV) / Incidence (EPV): the share of incidents reported to police. Victimization is nearly identical across the two localities, yet recorded crime is roughly twice as high in Barrios Unidos — the gap is driven by the reporting behavior, not the amount of crime.

Figure A.2: Information Source most Relevant to Assign Patrols and Allocate Resources



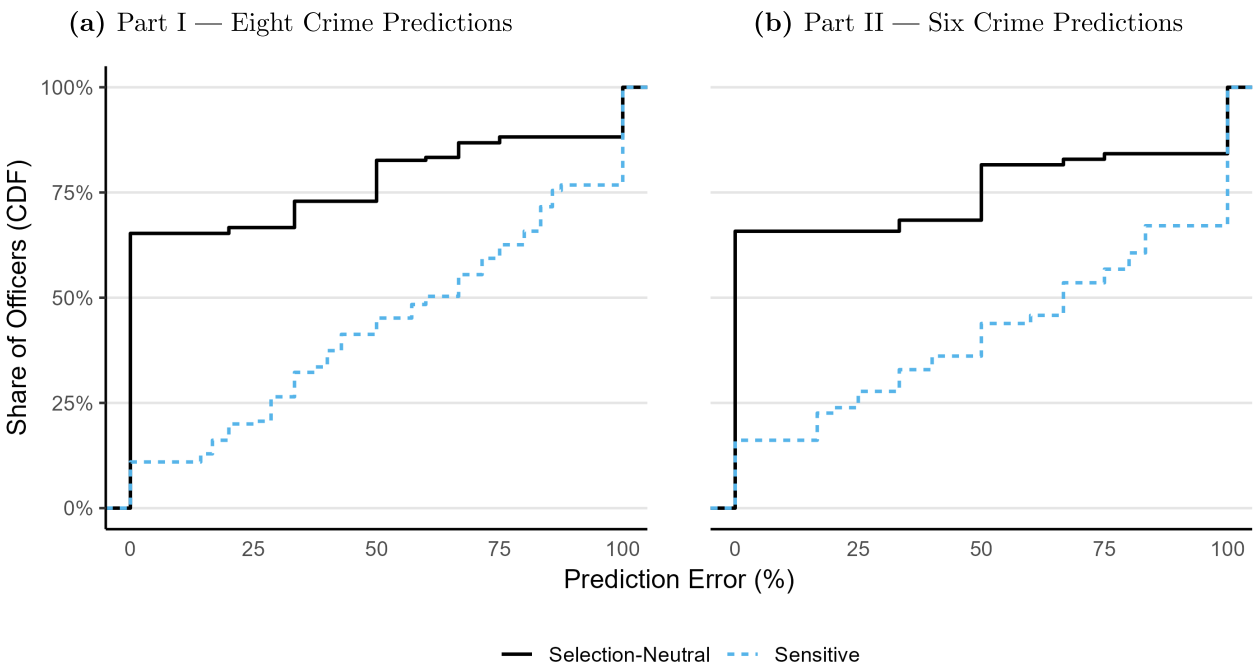
Notes: This figure plots a series of information sources and the share of officers who consider each of them as the most relevant to assign patrols and allocate resources in the field. Observations are one response per officer (N = 155).

Figure A.3: Map of reporting rates by locality, Bogotá (2025)



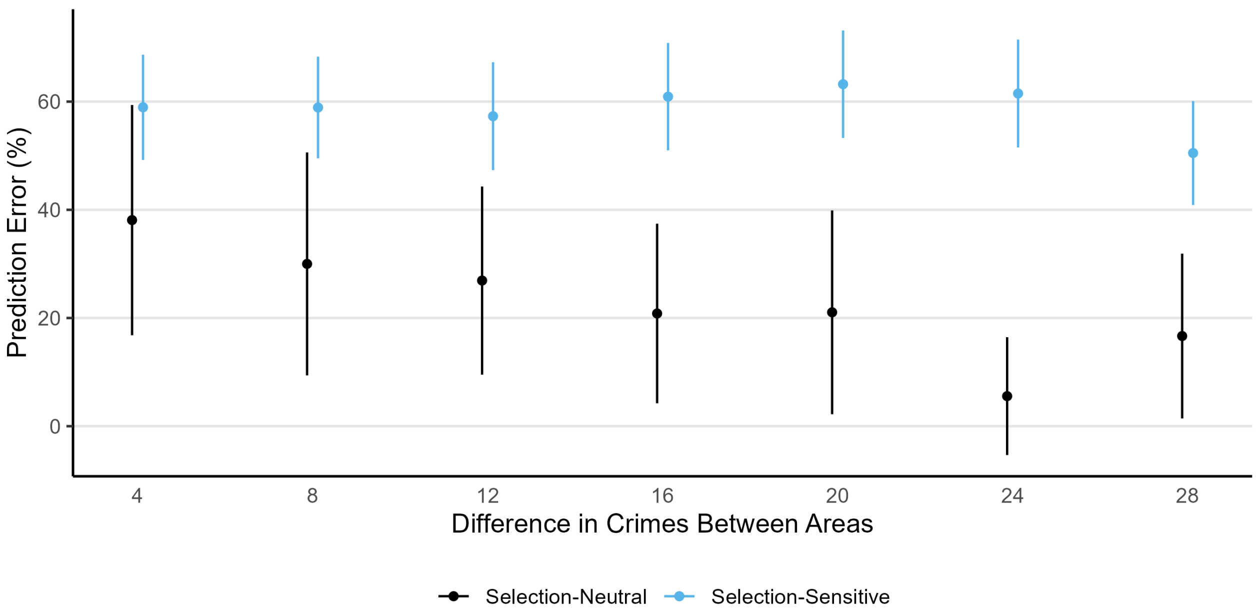
Notes: This map shows the reporting rate —the share of total crimes that were reported to the police— across 19 localities in Bogotá, according to the 2025 Perception and Victimization Survey of Bogotá (EPV, 2025). The survey asked over twenty thousand residents of Bogotá whether they had been the victim of a crime during 2025. Almost three thousand residents report at least one incident, for a total of 4,403 incidents. Out of these incidents, 43.5% were reported to the police.

Figure A.4: Distribution of Prediction Error



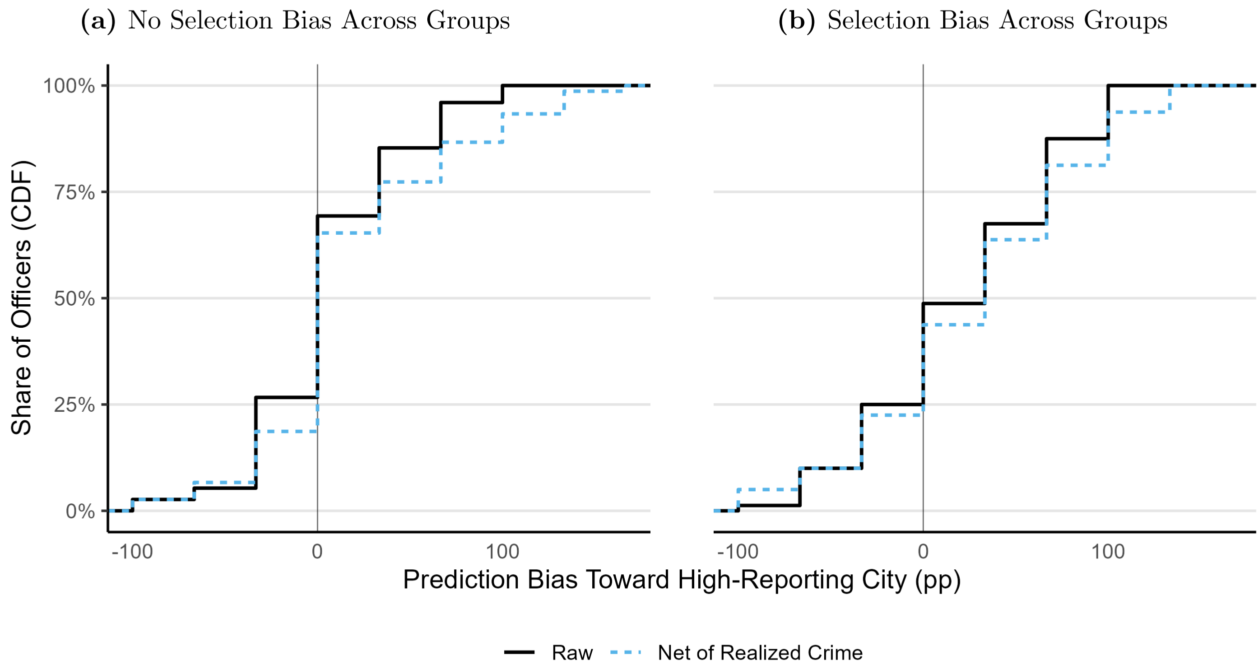
Notes: This figure shows the cumulative distribution of officer-level prediction error by type of prediction. Both panels divide the data by whether adjusting for selection was needed to make the right prediction. Selection-neutral decisions, where adjusting for selection does not change the optimal prediction, are depicted in black; selection-sensitive decisions, where adjusting for selection does change the optimal prediction, are in blue dashed lines. The left panel uses all eight predictions from Part I, the right panel uses the first six predictions from Part II.

Figure A.5: Prediction Error by Type of Prediction and Difference in Crimes



Notes: This figure disaggregates Figure 4 by the difference in crimes between areas at each prediction. The figure further divides the data by whether adjusting for selection was needed to make the accurate prediction. Selection-neutral decisions, where adjusting for selection does not change the optimal prediction, are depicted in black; selection-sensitive decisions, where adjusting for selection does change the optimal prediction, are depicted in blue. Error bars in both panels represent the 95% confidence interval of the mean, with standard errors clustered at the officer level.

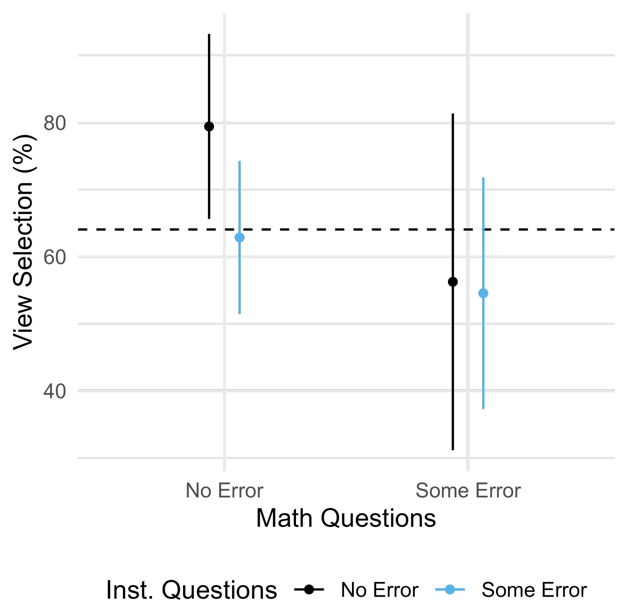
Figure A.6: Distribution of Bias Across Groups



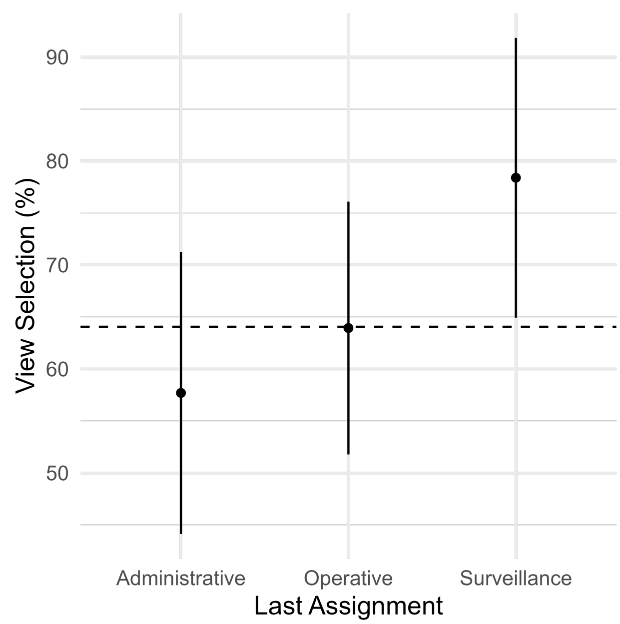
Notes: This figure shows the cumulative distribution of officer-level prediction bias, divided by treatment. Both panels show two measures of this bias. The black solid line shows the Raw bias, that is, how much more or less likely each officer is to predict more crime in the city with higher reporting rates on average. A Raw bias of 0 indicates that the officer was equally likely to predict crime in either city, while a bias of 100 indicates that the officer always predicted more crime in the city where it was more observable on average. While in expectation both cities have the same level of crime, the empirical realizations show some variance at the individual level: it is possible that the optimal prediction pattern for an individual officer is to always predict more crime in a given city. To account for the empirical realizations, we construct another measure of prediction error. A prediction is labeled as +1 if the officer *wrongly* predicted more crime in the high-reporting city, as -1 if they *wrongly* predicted more crime in the low-reporting city, and 0 if their prediction was accurate. The dashed blue line reports the distribution of this measure of error, Net of Realized Crime, at the officer level. The left panel restricts the sample to the officers in the No Selection Bias Across Groups treatment (75 officers), and the right panel to officers in the Selection Bias Across Groups treatment (80 officers). Observations are officer-level means, built from the first six predictions of Part II (left panel N = 75 officers $\times$ 6 predictions = 450; right panel N = 80 officers $\times$ 6 decisions = 480).

Figure A.7: Choice to View Selection

(a) By performance in math and instructions questions

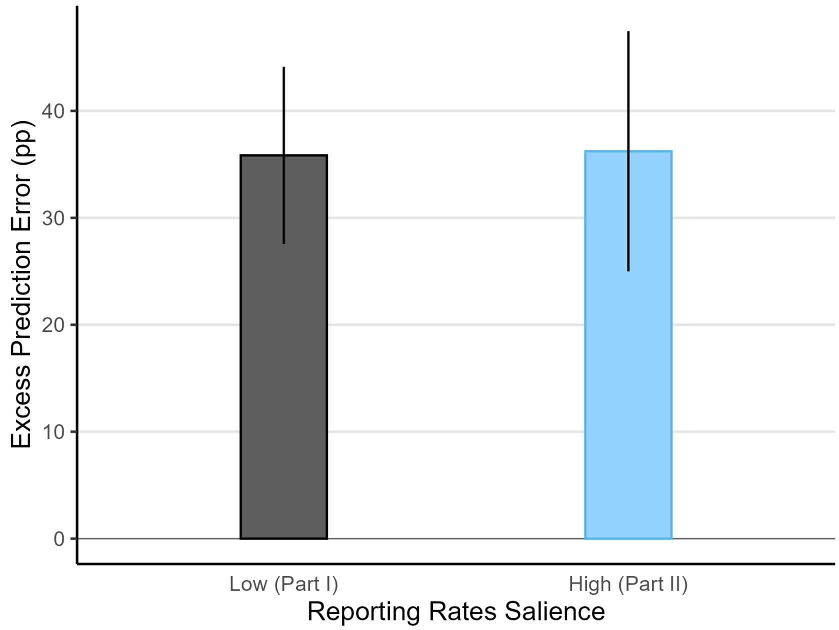


(b) By last assignment



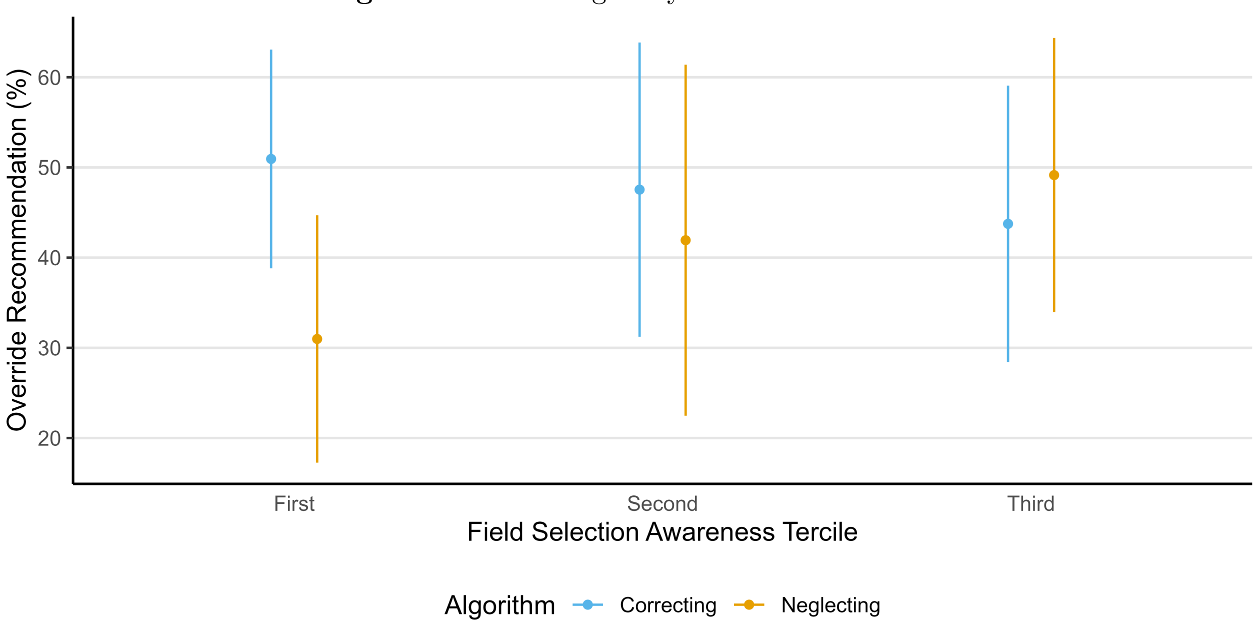
Notes: This figure shows the share of officers who choose to view selection for the last prediction of Part I. Panel (a) divides the sample by performance in the math and instruction comprehension questions. The left column shows officers who answered all math questions correctly, while the right column shows those who made at least one error. In black, we show officers who answered all instruction comprehension questions correctly, while in blue we show those who made at least one error. Panel (b) divides the sample by the last assignment of the officer before entering the academy. Note that those with a Vigilance assignment were in charge of predicting crime and assigning patrols. In both panels, error bars represent the 95% confidence interval around the mean. The dashed line shows the sample-level mean at 64%.

Figure A.8: Excess Prediction Error caused by Neglect, by Salience of Reporting Rates



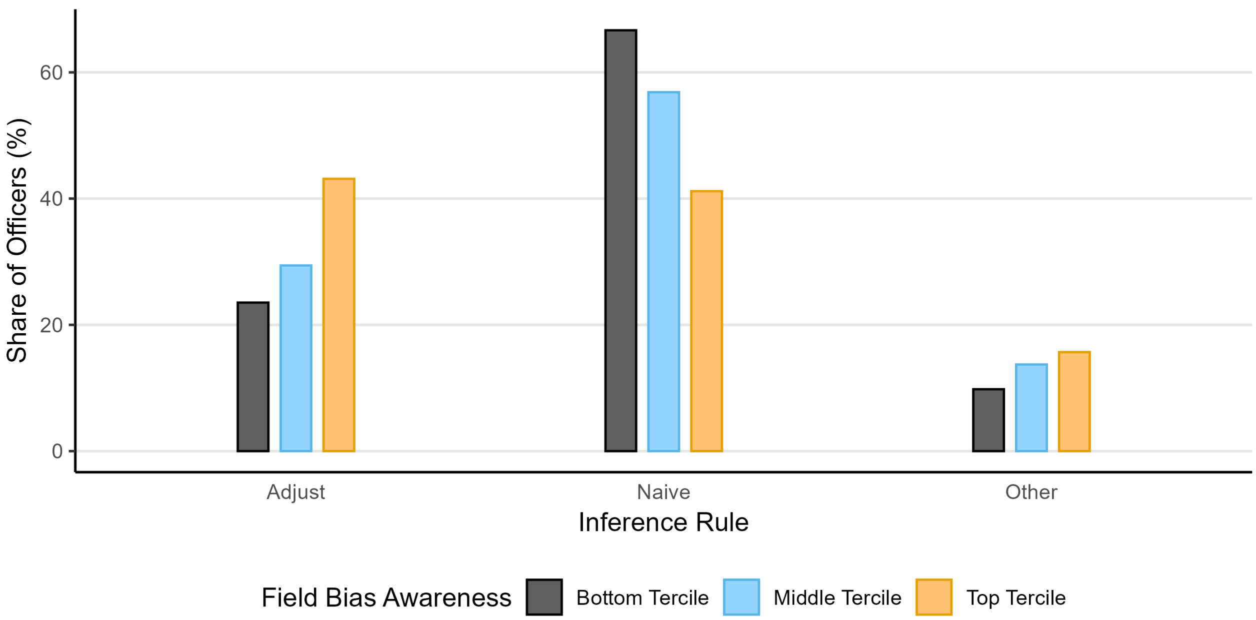
Notes: The figure shows the excess prediction error caused by selection neglect, estimated separately for Part I and Part II of the experiment, which vary in how salient reporting rates are. In particular, we regress prediction error on whether the prediction is selection-sensitive, separately for each part. The bars represent the coefficient of selection-sensitivity, along with error bars that show its 95% confidence interval with standard errors clustered at the officer level. Observations are officer-prediction pairs (Part I: N = 155 officers × 8 predictions = 1,240; Part II: N = 155 officers × 6 predictions = 930), from the middle eight predictions of Part I and the first six predictions of Part II, respectively.

Figure A.9: Heterogeneity in Advice Override



Notes: This figure shows the share of selection-sensitive decisions in which officers override the algorithm recommendation, dividing the sample in terciles according to their field selection awareness. Error bars show the 95% confidence interval of the mean, with standard errors clustered at the officer level.

Figure A.10: Inference Rules



Notes: This figure shows the inference rules that best capture officers' predictions in the first six rounds of Part I. Adjust reflects predicting more crime in the area with the higher number of reports adjusted by the reporting rate, Naive reflects taking the number of reports at face value, and Other reflects other strategies. We classify officers according to their first six predictions in Part II, and assign them to Adjust or Naive if one of these rules explains more than two thirds of their predictions, and to Other otherwise.

Table A.1: Patrolling time and reported crimes across eight police *cuadrantes*

	Patrol Time (sec)				
	Cross-section	Daily panel		Weekly panel	
	(1)	(2)	(3)	(4)	(5)
Reported Crimes (month)	0.308				
	(0.000)				
Reported Crimes, day t-1		0.671	0.015		
		(0.000)	(0.826)		
Reported Crimes, days t-1 to t-7				0.536	0.002
				(0.000)	(0.966)
Fixed-effects					
Cuadrante	Yes	Yes		Yes	
Cell			Yes		Yes
Observations	150	4,350	4,350	3,450	3,427
Dependent variable mean	61,031	2,012	2,012	1,852	1,864

Notes: Results from Poisson pseudo-maximum likelihood (PPML) regressions with $p-$values in parentheses and Conley (1999) spatial standard errors with a 1 km cutoff. The unit of observation is a 300m $\times$ 300m grid cell in column (1) and a cell $\times$ day in columns (2)-(5). The study area is the union of the Centenario Olaya and Ismael Perdomo CAI polygons, buffered by one cell width; the two cells containing the station buildings are excluded throughout. The dependent variable Patrol Time is the time patrolling officers spent in the cell, built from GPS pings weighted by the seconds elapsed until that officer's next fix, capped at 300 seconds. Reported Crimes counts SIEDCO crime reports —including personal injury, theft from persons, vehicle theft, motorcycle theft, and homicide (218 in total). We include Reported Crimes recorded in the cell over the month in column (1), on day $t-1$ in columns (2)-(3), and cumulatively over days $t-1$ to $t-7$ in columns (4)-(5). The panel covers June 2025; columns (2)-(3) drop the first day and columns (4)–(5) the first seven, for which the lags are undefined, and the cell fixed effects in column (5) absorb one cell with no patrol time in the estimation window. *Cuadrante* fixed effects correspond to the eight official patrol beats (three in Centenario, five in Perdomo) to which officers are permanently assigned; Cell fixed effects to the 150 grid cells. Coefficients are semi-elasticities: $\exp(\beta)-1$ is the proportional change in patrol time associated with one additional reported crime.

Table A.2: Correlates of reporting behavior in Bogotá

	Reported to Police
	(1)
Age	-0.002
	(0.001)
SES (Stratum)	0.033
	(0.001)
Female	0.003
	(0.854)
Education	0.021
	(0.000)
Type of crime (ref. Theft - persons)	
Cybercrime	0.068
	(0.005)
Extortion	0.095
	(0.036)
Personal injury	-0.004
	(0.901)
Sexual violence	-0.159
	(0.001)
Theft - bikes	-0.015
	(0.703)
Theft - commerce	-0.099
	(0.017)
Theft - residences	0.057
	(0.121)
Theft - vehicles	0.054
	(0.085)
Threats	-0.023
	(0.545)
Vandalism	-0.118
	(0.002)
Violence against women	-0.030
	(0.479)
Fixed Effects	
Locality	Yes
Occupation	Yes
Observations	4,403
Dependent variable mean	0.435

Notes: Results from a linear regression with $p-$values in parentheses and robust standard errors. The number of observations corresponds to 2,956 respondents who were the victim of a crime, out of 20,038 total respondents, with one row per crime suffered by the respondent. The dependent variable Reported to Police is a dummy that equals 1 if the respondent reported that crime to the police, conditional on being a victim. Age is measured in years. SES (Stratum) reflects the socio-economic stratum of the respondent, from 1 (Low) to 6 (High). Female is a dummy that equals 1 if the participant identifies as a woman. Education is an ordered factor variable from 1 (None) to 11 (PhD). Type of crime variables are dummies, taking as reference Theft - persons. The model includes Locality (20) $\times$ Respondent Occupation (8) fixed effects.

Table A.3: Prediction error across decisions

	Prediction Error					
		Part I			Part II	
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.221	0.327	0.159	0.230	0.333	0.136
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.006)
Selection-Sensitive	0.358	0.282	0.495	0.362	0.323	0.509
	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)	(0.000)
View Selection		-0.165			-0.155	
		(0.007)			(0.111)	
Selection-Sensitive × View Selection		0.119			0.047	
		(0.183)			(0.704)	
Calculator			0.011			0.103
			(0.867)			(0.310)
Selection-Sensitive × Calculator			-0.299			-0.427
			(0.002)			(0.002)
Observations	1,240	1,224	784	930	918	588
Dependent variable mean	0.493	0.493	0.468	0.533	0.528	0.485

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Prediction Error is a dummy that equals 1 if the prediction of which area had more crime was wrong. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. View Selection is a dummy that equals 1 if the officer chose to view the reporting rate for the last decision of Part I. Calculator is a dummy that equals 1 if that officer ever used the on-screen calculator through that part. Columns (1) to (3) use the eight predictions from Part I, with Column (3) restricting the sample to those who chose to view selection in the last decision of Part I. Columns (4) to (6) use the first six predictions in Part II. Column (6) restricts the sample to those who chose to view selection in the last decision of Part I.

Table A.4: Effect of crime and reporting gaps in prediction probability

	Predict Left					
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.469			0.499		
	(0.000)			(0.000)		
Crime Gap	0.002	0.003	0.001	0.003	0.003	-0.001
	(0.000)	(0.000)	(0.098)	(0.011)	(0.007)	(0.674)
Reporting Gap	0.003	0.004	0.003	0.003	0.003	0.003
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.008)
View Selection × Crime Gap			0.002			0.006
			(0.062)			(0.009)
View Selection × Reporting Gap			0.000			0.000
			(0.700)			(0.980)
Fixed-effects						
Officer	No	Yes	Yes	No	Yes	Yes
Observations	1,224	1,224	1,224	918	918	918
Dependent variable mean	0.469	0.469	0.469	0.487	0.487	0.487

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Predict A is a dummy that equals 1 if the area from city A was predicted to have higher crime (Columns (1) to (3)). The dependent variable Predict C is a dummy that equals 1 if the area from city C was predicted to have higher crime, using predictions from Part II (Columns (4) to (6)). Crime Gap is the difference in crimes between the two presented areas at each prediction. Reporting Gap is the difference in reporting rates between the two presented areas at each prediction, in percentage points. View Selection is a dummy that equals 1 if the officer chose to view the reporting rate for the last prediction of Part I. Columns (1) to (3) use the eight predictions of Part I, including the gap in demographic decoys as controls. Columns (4) to (6) use the first six predictions in Part II. Columns (2), (3), (5), and (6) include officer fixed effects.

Table A.5: Effect of all observable variables in prediction probability

	Predict Left					
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.467			0.494		
	(0.000)			(0.000)		
Δ Crime Reports	0.003	0.005	0.003	0.003	0.004	0.000
	(0.000)	(0.000)	(0.034)	(0.026)	(0.002)	(0.950)
Reporting Gap	-0.003	-0.006	-0.003	-0.005	-0.008	0.004
	(0.088)	(0.002)	(0.325)	(0.178)	(0.026)	(0.434)
Δ Age	-0.001	-0.001	-0.003			
	(0.422)	(0.531)	(0.279)			
Δ Unempt.	0.010	0.008	0.021			
	(0.000)	(0.001)	(0.000)			
Δ SES	0.022	0.017	0.037			
	(0.005)	(0.032)	(0.012)			
View Selection × Δ Crime Reports			0.003			0.007
			(0.105)			(0.008)
View Selection × Reporting Gap			-0.006			-0.018
			(0.159)			(0.008)
View Selection × Δ Age			0.003			
			(0.472)			
View Selection × Δ Unempt.			-0.020			
			(0.000)			
View Selection × Δ SES			-0.031			
			(0.067)			
Fixed-effects						
Officer	No	Yes	Yes	No	Yes	Yes
Observations	1,224	1,224	1,224	918	918	918
Dependent variable mean	0.469	0.469	0.469	0.487	0.487	0.487

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Predict A is a dummy that equals 1 if the area from city A was predicted to have higher crime (Columns (1) to (3)). The dependent variable Predict C is a dummy that equals 1 if the area from city C was predicted to have higher crime, using predictions from Part II (Columns (4) to (6)). Δ Crime Reports is the difference in crime reports between the two presented areas at each prediction. Reporting Gap is the difference in reporting rates between the two presented areas at each prediction, in percentage points. Δ Age is the difference in mean age between the two presented areas at each prediction. Δ Unemployment is the difference in unemployment rates between the two presented areas at each prediction, in percentage points. Δ SES is the difference in socioeconomic status (*estrato*, measured from 1 to 6) between the two presented areas. View Selection is a dummy that equals 1 if the officer chose to view the reporting rate for the last prediction of Part I. Columns (1) to (3) use the eight predictions of Part I. Columns (4) to (6) use the first six predictions in Part II. Columns (2), (3), (5), and (6) include officer fixed effects.

Table A.6: Prediction Error caused by neglect, robustness

	Prediction Error					
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.223	0.177	0.236	0.240	0.311	0.230
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Selection-Sensitive	0.291	0.328	0.296	0.304	0.226	0.320
	(0.000)	(0.000)	(0.000)	(0.000)	(0.030)	(0.000)
Some Math Error	-0.006			-0.025		
	(0.921)			(0.790)		
Selection-Sensitive × Some Math Error	0.211			0.178		
	(0.020)			(0.166)		
Some Inst. Error		0.070			-0.115	
		(0.221)			(0.248)	
Selection-Sensitive × Some Inst. Error		0.036			0.197	
		(0.685)			(0.114)	
Only Neutral Framing Correct	N	N	Y	N	N	Y
Observations	1,240	1,240	936	930	930	702
Dependent variable mean	0.493	0.493	0.460	0.533	0.533	0.499

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Prediction Error is a dummy that equals 1 if the prediction of which area had more crime was wrong. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Some Math Error is a dummy that equals 1 if the officer didn't answer all math questions correctly. Some Instruction Error is a dummy that equals 1 if the officer didn't answer all instruction comprehension questions correctly on the first attempt. Columns (3) and (6) restrict the sample to those officers who responded correctly to the neutral framing prediction question (75% of the sample). Columns (1), (2), and (3) use the eight predictions from Part I. Columns (4), (5), and (6) use the first six predictions in Part II.

Table A.7: Prediction Bias Across Groups

	Raw Gap			Net Gap		
	(1)	(2)	(3)	(4)	(5)	(6)
No Bias Across Groups	4.889	-2.564	1.282	16.889	12.821	11.538
	(0.301)	(0.749)	(0.831)	(0.007)	(0.161)	(0.122)
Bias Across Groups	20.000	6.944	10.897	26.667	19.444	17.308
	(0.001)	(0.449)	(0.112)	(0.000)	(0.119)	(0.031)
Sample	All	Inst. OK	Math OK	All	Inst. OK	Math OK
Observations	930	300	624	930	300	624
Dependent variable mean	12.688	2	6.090	21.935	16	14.423

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The regression does not include an intercept. The dependent variable Raw Gap shows how much more likely each officer is to predict more crime where it is more observable: P(High-reporting city) - P(Low-reporting city). The dependent variable Net Gap adjusts the latter by the empirical optimal prediction, so a prediction is labeled as +1 if the officer wrongly predicted more crime in the high-reporting city, as -1 if they wrongly predicted more crime in the low-reporting city, and 0 if their prediction was accurate. No (Selection) Bias Across Groups is a dummy that equals 1 if the officer was assigned to the treatment where cities have equal reporting rates in expectation, while (Selection) Bias Across Groups marks the treatment where reporting rates are systematically higher in one city. Because the specification includes no intercept, each coefficient reflects the average prediction gap between the high-reporting city and the low-reporting city for the corresponding treatment. Columns (1) and (4) use the entire sample (N = 155 officers $\times$ 6 predictions), while Columns (2) and (5) restrict the sample to those officers who made no mistake in the instruction comprehension questions (N = 50 officers $\times$ 6 predictions), and Columns (3) and (6) restrict the sample to those officers who made no mistake in the arithmetic questions (N = 104 officers $\times$ 6 predictions).

Table A.8: Heterogeneity in choice to view reporting rates

	View Reporting Rates			
	(1)	(2)	(3)	(4)
Constant	0.762	0.577	0.649	0.708
	(0.000)	(0.000)	(0.000)	(0.000)
Some Math Error	-0.132			-0.136
	(0.125)			(0.107)
Some Instruction Error	-0.118			-0.135
	(0.139)			(0.087)
Operative		0.062		0.076
		(0.503)		(0.412)
Surveillance		0.207		0.193
		(0.035)		(0.051)
Field Bias Awareness			0.107	0.110
			(0.212)	(0.181)
Observations	153	150	151	150
Dependent variable mean	0.641	0.653	0.649	0.653

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable View Reporting Rates is a dummy that equals 1 if the officer chose to view the reporting rates for the tenth decision of Part I. Some Math Error is a dummy that equals 1 if the officer didn't answer all math questions correctly. Some Instruction Error is a dummy that equals 1 if the officer didn't answer all instruction comprehension questions correctly on the first attempt. Operative is a dummy that equals 1 if the officer's last assignment was operative. Surveillance is a dummy that equals 1 if the officer's last assignment was in surveillance —predicting crime and assigning patrols. The reference group is officers whose last assignment was administrative. Field Bias Awareness measures the officer's perception of the biases in crime reports in the field, with higher values implying higher skepticism about this data. All columns use data at the officer level, as there is only one choice to view selection.

Table A.9: Prediction error due to neglect, heterogeneity by assignment and field bias awareness

	Prediction Error					
		Part I			Part II	
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.212	0.218	0.212	0.264	0.233	0.256
	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)
Selection-Sensitive	0.428	0.358	0.426	0.373	0.362	0.378
	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)
Operative	-0.002		-0.007	-0.064		-0.060
	(0.971)		(0.917)	(0.528)		(0.523)
Surveillance	0.038		0.050	-0.037		-0.005
	(0.609)		(0.465)	(0.751)		(0.962)
Selection-Sensitive × Operative	-0.131		-0.118	-0.015		-0.008
	(0.168)		(0.208)	(0.916)		(0.952)
Selection-Sensitive × Surveillance	-0.082		-0.104	-0.013		-0.056
	(0.485)		(0.343)	(0.933)		(0.695)
Bias Awareness		0.115	0.124		0.236	0.238
		(0.082)	(0.068)		(0.006)	(0.006)
Selection-Sensitive × Bias Awareness		-0.255	-0.262		-0.388	-0.397
		(0.010)	(0.008)		(0.001)	(0.001)
Observations	1,216	1,224	1,216	912	918	912
Dependent variable mean	0.488	0.489	0.488	0.532	0.535	0.532

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Prediction Error is a dummy that equals 1 if the prediction of which area had more crime was wrong. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Operative is a dummy that equals 1 if the officer's last assignment was operative. Surveillance is a dummy that equals 1 if the officer's last assignment was in surveillance —predicting crime and assigning patrols. The reference group is officers whose last assignment was administrative. (Field) Bias Awareness measures the officer's perception of the biases in crime reports in the field, with higher values implying higher skepticism about this data. It's built by aggregating the z-scores of the five questions about biases in crime data in the field, following Kling et al. (2007). Columns (1) to (3) use the eight predictions from Part I. Columns (4) to (6) use the first six predictions in Part II.

Table A.10: Standardized response times by decision features

	Standardized Response Time			
	Part I		Part II	
	(1)	(2)	(3)	(4)
Constant	-0.049	0.272	-0.062	0.298
	(0.515)	(0.038)	(0.509)	(0.135)
Selection-Sensitive	0.064	-0.400	0.074	-0.444
	(0.317)	(0.003)	(0.438)	(0.032)
Accurate		-0.411		-0.468
		(0.003)		(0.025)
Selection-Sensitive × Accurate		0.752		0.855
		(0.000)		(0.001)
Observations	1,240	1,240	930	930
Dependent variable mean	0.000	0.000	0.000	0.000

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Standardized Response Time is the z-scored response time for each prediction. Accurate is a dummy that equals 1 if the prediction of which area had more crime was accurate. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Columns (1) and (2) use the eight predictions from Part I. Columns (3) and (4) use the first six predictions of Part II.

Table A.11: Algorithmic advice override and prediction error

	Override		Prediction Error	
	Low Neglect	High Neglect	Low Neglect	High Neglect
	(1)	(2)	(3)	(4)
Constant	0.604	0.218	0.364	0.318
	(0.000)	(0.000)	(0.001)	(0.000)
Correcting	-0.284	0.449	-0.078	-0.170
	(0.002)	(0.000)	(0.543)	(0.071)
Selection-Sensitive			0.013	0.493
			(0.920)	(0.000)
Neglecting			-0.137	-0.244
			(0.302)	(0.003)
Selection-Sensitive × Correcting			0.022	0.153
			(0.890)	(0.264)
Selection-Sensitive × Neglecting			0.080	0.333
			(0.642)	(0.003)
Observations	191	179	426	504
Dependent variable mean	0.455	0.413	0.319	0.593

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Override is a dummy that equals 1 if the participant overrides the algorithmic advice for that prediction. The dependent variable Prediction Error is a dummy that equals 1 if the prediction of which area had more crime was wrong. Correcting is a dummy that equals 1 if the officer was assigned to the Correcting Algorithm condition. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Neglecting is a dummy that equals 1 if the officer was assigned to the Neglecting Algorithm condition. Columns (1) and (2) only use the selection-sensitive predictions within the last six rounds of Part II, as both the correcting and the neglecting algorithm give the same advice in selection-neutral ones, and do not include officers in the Control condition, as they didn't receive any advice. Columns (3) and (4) use the last six predictions of Part II. Columns (1) and (3) restrict the sample to those officers who had a below-median error rate in selection-sensitive predictions in the first six predictions of Part II (before the algorithm is introduced), Columns (2) and (4) to its complement. Note that the reference group in Columns (1) and (2) is the Neglecting algorithm condition, while in (3) and (4) is the Control condition.

B Data Generating Process

The number of crimes in the two areas of each prediction is generated jointly. We first draw crime in one area from ($c_{1n} \sim \mathcal{N}(250, 20^2)$), rounded to the nearest integer. We then set crime in the other area to ($c_{2n} = c_{1n} + \varepsilon_n$), where the crime gaps (ε_n) are drawn without replacement from a symmetric set centered at zero. City labels are randomly assigned, ensuring that crime has the same distribution across cities and experimental conditions. Participants do not know the exact distribution. Reporting rates (s_{gn}) are drawn independently from the uniform distributions corresponding to each experimental condition and take values in multiples of five percentage points to ease computation. We then deterministically construct crime reports as ($r_{gn} = \text{round}(c_{gn} s_{gn})$). Reported crime is therefore a selected signal of underlying crime, as it depends on both crime and its observability. Adjusting for selection requires dividing reports by the reporting rate, (r_{gn}/s_{gn}). By construction, rounding never changes the ranking of crime across areas, so a decision-maker who correctly adjusts for selection always makes the correct prediction. Prediction errors therefore reflect cognitive error rather than statistical noise. In Part I, we also include demographic characteristics as decoys that reduce the salience of reports and reporting rates. These variables—average age, unemployment, and socioeconomic status (SES)[9]—are drawn independently from uniform distributions and are, by construction, uninformative about crime.

Selection Bias Across Groups — All features of the DGP are identical except that the reporting rates for one city are drawn from $\mathcal{U}[0.1, 0.6]$ and for the other are drawn from $\mathcal{U}[0.4, 0.9]$, instead of both being drawn from $\mathcal{U}[0.1, 0.9]$. Which city gets assigned the higher reporting rates is a coin flip.

[9]In Colombia, residential areas are classified into socioeconomic strata (*estrato*) ranging from 1 (low) to 6 (high). This classification is used to set utility prices and target public services and is a familiar proxy for income.

C Sample Restriction

Why excluding some participants

Interpreting prediction errors as selection neglect requires participants to engage with the basic information presented in the task. Our preregistration anticipated excluding participants whose choices were inconsistent with the data shown to them. Accordingly, the first prediction serves as a minimum task-engagement check. Participants see only the number of crime reports in each area, with no information about reporting rates. Given the experimental data-generating process, the prescribed prediction is therefore the area with more reports. The restriction does not directly select participants according to whether they adjust for selection: the check contains no reporting-rate information and requires no selection adjustment. It also leaves little researcher discretion, as it consists of a single predetermined binary decision with one prescribed answer. 27 of the 182 participants fail this check.

Participants who fail the check behave differently throughout the experiment in ways consistent with lower task engagement. Table A.12 shows that, relative to participants who pass, they spend 0.34 standard deviations less time on each prediction in Part I ($p = 0.025$) and 0.57 standard deviations less time on each prediction in the main task ($p < 0.001$). Their prediction errors are also indistinguishable from chance. Their error rates in selection-neutral decisions are 47.8% in Part I ($p = 0.797$; $H_0 = 50\%$) and 60.0% in the main task ($p = 0.450$; $H_0 = 50\%$). These decisions require no selection adjustment, as following reports already produces the correct prediction.

One possible concern is that the check excludes especially bias-aware officers who deliberately distrust crime reports. The subsequent decisions do not support this interpretation. Participants who fail the check are also no more accurate than chance in selection-sensitive predictions (43.9%, $p = 0.419$; $H_0 = 50\%$), where systematically predicting more crime in the area with fewer reports would improve accuracy. They also do not differ from the rest of the sample in field bias awareness ($p = 0.759$). Taken together, their shorter response times and absence of a relationship between the displayed information and their predictions are more consistent with limited task engagement than with deliberate correction for selection.

Are our results driven by the sample restriction?

Our results are robust to including the dropped participants in the analysis. Table A.13 reports the main results separately for participants who pass the check, the full sample, and participants who fail the check. Including all participants attenuates the estimated magnitudes but preserves the sign and statistical significance of the headline results. The excess error in selection-sensitive decisions falls from 36.2 to 27.9 percentage points; the reporting-gap coefficient falls from 0.348 to 0.281; and the group prediction gap in the Selection Bias Across Groups condition falls from 20.0 to 15.9 percentage points.

The estimates among participants who fail the check illustrate why their predictions have no clear interpretation as selection neglect. Their errors do not differ between selection-sensitive and selection-neutral decisions and their predictions do not respond to reporting-rate gaps. Including these participants therefore combines errors attributable to selection neglect with behavior that is mostly consistent with random responses.

Table A.12: Characteristics of officers included and dropped by the sanity check

	Included (1)	Dropped (2)	Difference (2)−(1)
Age	36.1	37.1	1.0
			(0.095)
Male	0.892	0.704	−0.188
			(0.009)
Years in the National Police	13.9	14.8	0.9
			(0.153)
All comprehension questions correct	0.323	0.185	−0.137
			(0.153)
All arithmetic questions correct	0.671	0.481	−0.189
			(0.058)
Field bias awareness (SD)	−0.004	0.025	0.030
			(0.759)
Response time, Part I (SD)	0.115	−0.226	−0.341
			(0.025)
Response time, main task (SD)	0.151	−0.421	−0.572
			(<0.001)
Officers	155	27	

Notes: The sanity check is the first prediction of Part I, where officers see only crime reports and should predict the area with more of them. Columns (1) and (2) report means for officers who pass and fail it. Column (3) is the difference, with the p-value of a two-sample test in parentheses. Field bias awareness is the index of five survey items, signed so that higher values mean greater recognition that reported crime data is a biased sample, in standard deviations. Response times are officer-level means, standardised over the pooled sample, for the eight predictions in Part I and the first six predictions of Part II.

Table A.13: Main results by sample

	Main (1)	Full (2)	Dropped (3)
Panel A. Prediction error (%)			
Selection-neutral	23.0	29.1	60.0
	(4.3)	(4.5)	(13.2)
Selection-sensitive	59.3	57.0	43.9
	(3.0)	(2.8)	(7.5)
Sensitive − neutral	36.2	27.9	−16.1
	(5.7)	(5.8)	(17.2)
Panel B. Prediction bias (pp)			
Reporting-gap slope (pp per pp)	0.348	0.281	−0.080
	(0.076)	(0.071)	(0.190)
Group gap: no selection bias	4.9	5.2	6.7
	(4.7)	(4.2)	(9.7)
Group gap: selection bias	20.0	15.9	−11.1
	(5.8)	(5.7)	(19.4)
Panel C. Information demand (%)			
View reporting rates	64.1	60.0	37.0
	(3.9)	(3.7)	(9.5)
Officers	155	182	27
Observations	930	1092	162

Notes: Column (1) is the sample used throughout the paper, officers who pass the sanity check. Column (2) adds the officers who fail it. Column (3) is those officers on their own. Standard errors clustered at the officer level in parentheses. Observations in Panels A and B are officer-prediction pairs, from the first six predictions of Part II; the information demand row in Panel C is one observation per officer. Panel A reports mean prediction error by prediction type and their difference. Among participants who pass the check, this gap is our primary measure of selection neglect. The reporting-gap slope is the coefficient of the prediction on the reporting-rate gap between areas, holding the crime gap and the officer fixed, in percentage points of prediction per percentage point of reporting gap. The group gap rows are the difference in prediction probability between the high- and low-reporting city, by condition.